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List of Symbols and abbreviations

Symbols are separated per chapter of occurrence. Symbols may appear in multiple Chapters once they have been defined.

Symbol	Description
BM	Brownian Motion
a	Particle radius
z	Colloid center to wall distance
ρ_{object}	Object density
g	Gravity acceleration
Δm	Buoyant mass
D_0	Bulk diffusion coefficient
$D_{\perp}$	Effective diffusion coefficient for a normal diffusion
$D_{\parallel}$	Effective diffusion coefficient for an in-plane diffusion
$D_{i\alpha,j\beta}$	Diffusion matrix component for colloids labeled i and j , in directions α and β
k_B	Boltzmann's constant
Θ	Temperature
l_B	Boltzmann length
l_D	Debye length
B	Adimensionalised electrostatic repulsion force
η	Fluid viscosity
$\eta_{\perp}$	Effective viscosity for a normal motion
$\eta_{\parallel}$	Effective viscosity for an in-plane motion
ξ	Brownian white noise, zero mean and unit variance
MSD	Mean Squared Displacement
PDF	Probability Density Function
η	Dynamic viscosity of the fluid
ρ	Fluid density
q_i	Degree of freedom representing colloid i
$\phi(q_1)$	Sinusoidal potential acting upon q_1
ΔU	Sinusoidal trap amplitude
λ	Sinusoidal trap spatial period
κ	Inverse stiffness or compliance
ϵ	Adimensionalised compliance

E	Elastic modulus
k_{eff}	Local trap stiffness around a minimum of $\phi(q_1)$
D_{m}	Microscopic diffusion coefficient of mode q_1
D_2	Microscopic diffusion coefficient of mode q_2
τ_i	Characteristic relaxation time of q_i
Δt	Simulation time step
T_{sim}	Total simulation time
n_{steps}	Total number of iteration inside the trajectory simulation loop
ε	Convergence criterion
$\sigma(x)^2$	Variance of x
γ	Friction matrix
μ	Mobility matrix
μ_{e}	Dressed mobility
$D^*(\kappa)$	Effective long-time diffusion coefficient
R_{e}	Reynolds number
U	Characteristic velocity
L	Characteristic length
P_{e}	Péclet number
$\bar{u}$	Mean crosswise fluid velocity
W	Channel half-width
H	Channel thickness
τ_{D}	Transverse diffusion time
D_{eff}	Dispersion coefficient
ΔP	Entry-to-exist pressure difference
$\dot{\gamma}$	Shear rate
D_y	Crosswise diffusion coefficient
D_y^∞	Crosswise diffusion coefficient away from wall interactions
N	Number of colloids
Φ	Volume fraction
τ	Time lag between TIRFM experiment frames
λ_{e}	Evanescent wave visualisation limit
D_{T}	Translational diffusion coefficient
D_{R}	Rotational diffusion coefficient
ω	Angular velocity
Ω	Angular velocity vector
ABP	Active Brownian Particle
RNT	Run and Tumble particle

Introduction

Chapter 1

Brownian motion

This manuscript presents the work I have conducted in the frame of my PhD thesis in Brownian motion in complex environments. The development is organised in four core Chapters presenting each a development I have contributed to on Brownian motion. This first Chapter, which counts a general Introduction to the rest of the manuscript, introduces the notion of Brownian motion from a historical standpoint. I intend to remain very broad in this Chapter, and provide then only the key developments for a simple Brownian colloid in an unbounded fluid. Brownian motion is not only relevant in the field of History of Sciences, but also in current research. I present some recent developments in this Chapter, but each Chapter also showcases recent key research. Finally, I provide a numerical model to generate Brownian trajectories and extract information accurately.

I identify as a numerician and use computational tools to conduct my research. Most of the codes written for this manuscript and attached publications can be found in a dedicated GitHub repository. I invite those who are not familiar with numerical tools to attempt recreating the results presented in my Thesis using these codes. I attribute a lot of worth to open science, and my contributions, thought humble, are fully free to build upon, as long as their remain in the public domain.

1.1 History of Brownian Motion

1.2 Theoretical developments

1.2.1 The Fokker-Plank equation

$$dX_t = u(X_t)dt + g dB_t , \quad (1.1)$$

where $u(X_t)$ is the drift velocity term, g is the magnitude of the noise, and dB_t is the increment of a Wiener process.

We now take the average value of a function obeying the PDF $P(X_t, X_0; t)$, which we will call $f(X_t)$. This average value will, by definition, follow [1]:

$$\langle f(X_t) \rangle = \int f(X_t) P(X_t, X_0; t) dX_t , \quad (1.2)$$

with the initial condition $P(X_t, X_0; 0) = \delta(X_t - X_0)$, where δ is the Dirac delta function. Expanding $f(X_t)$ in a Taylor series yields:

$$f(X_t + dX_t) = f(X_t) + f'(X_t)dX_t + \frac{1}{2}f''(X_t)(dX_t)^2 + \dots \quad (1.3)$$

Substituting dX_t from the Fokker-Plank equation and taking the average, we get:

$$\left\langle \frac{df(X_t)}{dt} \right\rangle \simeq \frac{\partial f(X_t)}{\partial X_t} u(X_t) + \frac{1}{2} \frac{\partial^2 f(X_t)}{\partial X_t^2} g^2 \quad (1.4)$$

Combining this with the definition of the average value of $f(X_t)$, we can derive the Fokker-Plank equation for the PDF $P(X_t, X_0; t)$:

$$\int dX_t \frac{\partial f(X_t)}{\partial t} f(X_t) = \frac{\partial f(X_t)}{\partial X_t} u(X_t) + \frac{1}{2} \frac{\partial^2 f(X_t)}{\partial X_t^2} g^2 \quad (1.5)$$

$$= \int dX_t P(X_t, X_0; t) G f(X_t) , \quad (1.6)$$

where G is the Generator, defined as $G = u(X_t) \frac{\partial}{\partial X_t} + \frac{1}{2} g^2 \frac{\partial^2}{\partial X_t^2}$. Taking the adjoint of G as $G^\dagger$, we can rewrite the above equation as:

$$\int dX_t P(X_t, X_0; t) G f(X_t) = \int dX_t f(X_t) G^\dagger P(X_t, X_0; t) . \quad (1.7)$$

We can then derive the Fokker-Plank equation for the PDF $P(X_t, X_0; t)$ as:

$$\frac{\partial P(X_t, X_0; t)}{\partial t} = G^\dagger P(X_t, X_0; t) = -\frac{\partial}{\partial X_t} (u(X_t) P(X_t, X_0; t)) + \frac{1}{2} \frac{\partial^2}{\partial X_t^2} (g^2 P(X_t, X_0; t)) . \quad (1.8)$$

Thus, for a bulk Brownian motion with Langevin equation

$$\frac{dx}{dt} = \sqrt{2D_0} \eta(t) , \quad (1.9)$$

with x the colloid's position, D_0 the bulk diffusion coefficient, and $\eta(t)$ a white noise of zero mean and unit variance, we can identify $u(x) = 0$ and $g = \sqrt{2D_0}$, and thus, the Fokker-Plank equation for the PDF of the colloid's position reads:

$$\frac{\partial P(x, x_0; t)}{\partial t} = D_0 \frac{\partial^2 P(x, x_0; t)}{\partial x^2} , \quad (1.10)$$

which is the well-known diffusion equation.

1.3 Advances and state of the Art

1.4 Bases of numerical simulations

Chapter 2

Appendix 1: Publications

Characterising anomalous diffusion with a state-space deep-learning model

Hydrodynamic couplings between a colloid trapped in a periodic potential and a soft and fluctuating membrane enhance long-time diffusion

Active particles advected by viscous fluid flows undergo Taylor-Aris dispersion

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PAPER

Bidirectional Mamba state-space model for anomalous diffusion

Maxime Lavaud^{1,2,*}, Yosef Shokeeb¹, Juliette Lacherez¹, Yacine Amarouchene^{1,*} and Thomas Salez^{1,*}¹ University Bordeaux, CNRS, LOMA, UMR 5798, Talence, F-33400, France² University Bordeaux, CNRS, Bordeaux INP, CBMN, UMR 5248, Pessac, F-33600, France

* Authors to whom any correspondence should be addressed.

E-mail: maxime.lavaud@u-bordeaux.fr, yacine.amarouchene@u-bordeaux.fr and thomas.salez@cnrs.fr**Keywords:** anomalous diffusion, deep learning, Mamba, particle tracking

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Abstract

Characterizing anomalous diffusion (AnDi) is crucial in order to understand the evolution of complex stochastic systems, from molecular interactions to cellular dynamics. Here, we evaluate the performances regarding such a task of Bi-Mamba, a novel state-space deep-learning architecture articulated around a bidirectional scanning mechanism. Our implementation is tested on the AnDi-2 challenge datasets. Designed for regression tasks, the Bi-Mamba architecture infers efficiently the effective diffusion coefficient and anomalous exponent from single, short trajectories. As such, our results indicate the potential practical use of the Bi-Mamba architecture towards the characterization of AnDi.

1. Introduction

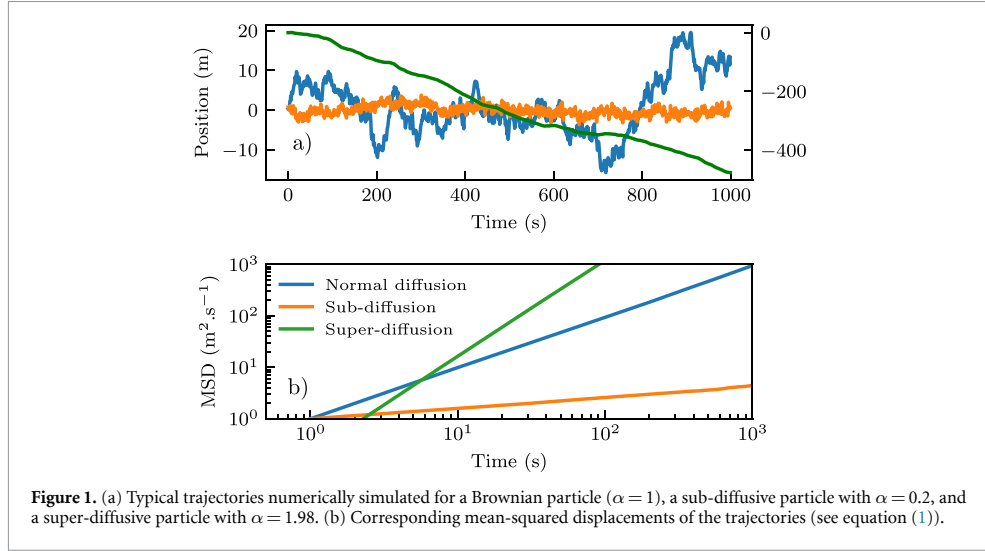
Anomalous diffusion (AnDi) [1] is ubiquitous in complex stochastic processes, and central in particular to the cell machinery, as classically shown by microrheology [2]. Its understanding provides insights on the transport of microparticles in heterogeneous media, such as molecules in biological cells [3, 4]. Methods such as microscopy combined with single-particle tracking allow for the detailed spatiotemporal analysis of single events [5, 6]. Over the last few decades, single-particle imaging methods have been steadily upgrading, increasing the amount of experimental data available on molecular interactions, cellular dynamics, microparticle behaviors [7, 8], and nanoparticles [9] in various complex media. Other approaches, such as dynamic light scattering [10, 11] and differential dynamic microscopy [12] have provided valuable insights on dense particle suspensions.

However, in most biological cases, trajectories are typically scarce, short and noisy. As such, it is often hard, if not impossible, to infer meaningful information using traditional tracking methods. Deep-learning methods for advanced microscopy have thus emerged as a promising change of paradigm [13]. As such, the work we present here results from the AnDi challenge [14], and in particular from our participation to its second edition [15]. We took a step aside from the standard methods used by the other participants in order to improve the collective knowledge on advanced artificial-intelligence methods for complex particle tracking. Our results indicate that the novel Bi-Mamba model infers efficiently the effective diffusion coefficient and anomalous exponent from single, short and noisy trajectories.

In a typical two-dimensional anomalous-diffusion process, the mean-squared displacement (MSD) is determined over time t by:

$$\text{MSD}(t) = 4Kt^\alpha, \quad (1)$$

where K is the effective diffusion coefficient and α is the AnDi exponent. For Brownian particles, $\alpha = 1$. Besides, a particle is in a sub-diffusive regime when $0 < \alpha < 1$, and in a super-diffusive regime when $1 < \alpha < 2$. In figure 1, typical samples of these different diffusion types are shown with their respective trajectories and MSDs. For example, in complex cell environments a particle can exhibit AnDi due to:



- *Obstacles and crowding*: in the presence of obstacles, diffusion can be hindered, leading to a sub-diffusive behavior. Moderate concentrations of obstacles can cause AnDi over short distances, and the effect becomes more pronounced as the obstacle concentration approaches the percolation threshold [16];
- *Binding and trapping*: temporary binding of particles to fixed sites [17] leads to a sub-diffusive behavior;
- *Lipid rafts and membrane heterogeneity*: interactions of particles with lipid rafts in cell membranes have been shown to lead to complex diffusive behaviors [18].

The first AnDi Challenge took place in 2020 and aimed at assessing the performances of various methods in quantifying AnDi [14]. The main focus was dedicated to the inference of K and α , for various simulations of bio-mimetic cases. The second AnDi challenge took place in 2024 and aimed at evaluating the performances of various methods for detecting and quantifying changes in single-particle motion [15]. As such, the focus was not only on the inference of the effective diffusion coefficient and exponent, but also on trajectory segmentation, with the trajectory statistics being described by a maximum of two K and two α values. One typical goal is for example to measure at what rate a particle binds and unbinds to/from a given cellular site. The AnDi datasets involve five phenomenological models:

- *Single-state*: particles diffusing according to a single diffusion state, as observed for some lipids in the plasma membrane [19–21];
- *Multi-state*: particles diffusing according to two diffusion states, and undergoing transient changes of K and α , as observed for proteins due to allosteric changes or ligand binding [22–25];
- *Dimerization*: particles diffusing according to two diffusion states, and undergoing transient changes of K and α , induced by encountering other diffusing particles, as observed in protein dimerization and protein-protein interactions [26, 27];
- *Transient confinement*: particles undergoing transient diffusion changes when entering or leaving given areas, as observed in the confinement induced by clathrin-coated pits on cell membranes [28];
- *Quenched trap*: two-state model of diffusion representing proteins being transiently immobilized at specific locations, induced by binding to immobile structures, as observed in cytoskeleton-induced molecular pinning [29, 30].

2. Bidirectional state-space model

Deep-learning models for inference from time series are generally based on recurrent neural networks (RNNs) or convolutional neural networks (CNNs). RNNs process the data sequence by updating a hidden state after each element. However, as the gradient also needs to go through each element one by one, long-term information may be lost. CNNs use convolutional kernels to give more weight to local information, as they do not need to process the sequence one by one. CNNs benefit from parallel computing and faster training speed but also limit the retrieval of long-term global information.

Algorithm 1. Bi-Mamba.

```

Input:  $x \in \mathbb{R}^{(B,L,D)}$ 
Output:  $y \in \mathbb{R}^{(B,L,D)}$ 
1  $x \leftarrow \text{Norm}(x)$ ;
2 for  $i \in \{\text{forward}, \text{backward}\}$  do
3   if  $i = \text{forward}$  then
4      $x_i \leftarrow x$ ;
5   else
6      $x_i \leftarrow \text{Flip}(x)$ ;
7   end
8    $x_i \leftarrow \text{MambaBlock}(x_i)$ ;
9   if  $i = \text{forward}$  then
10     $x_i \leftarrow \text{Flip}(x_i)$ ;
11  end
12   $x_i \leftarrow \text{Norm}(x_i + x)$ ;
13 end
14  $y_M = x_{\text{forward}} + x_{\text{backward}}$ ;
15  $y \leftarrow \text{FeedForward}(y_M)$ ;
16  $y \leftarrow \text{Norm}(y + y_M)$ ;
17 return  $y$ ;

```

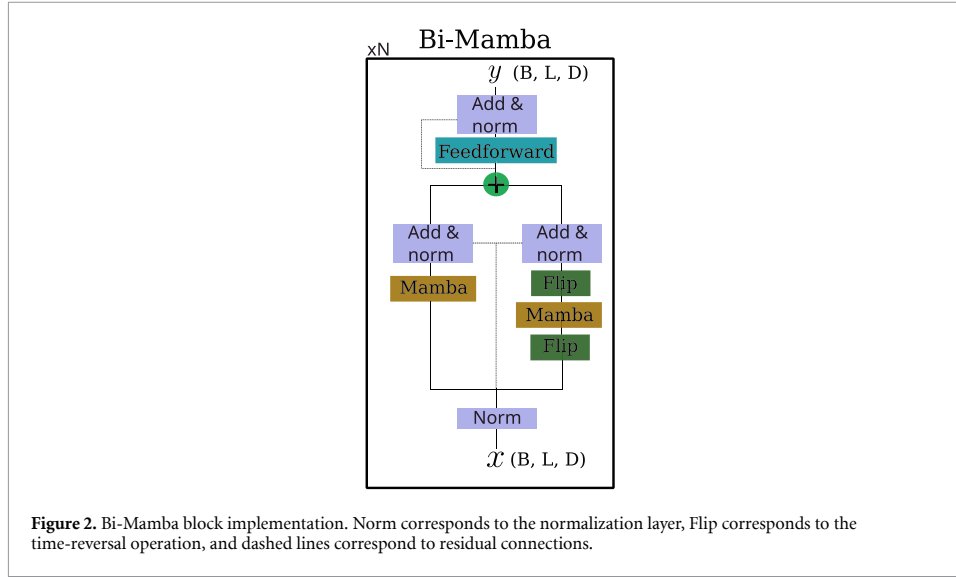
The Mamba architecture [31] is a state-space model designed to overcome the limitations of traditional deep-learning models like RNNs and CNNs by handling long-term dependencies with a high training efficiency. The key innovations of the Mamba architecture include a selection mechanism and hardware optimization for GPU usage. The Mamba architecture is essentially a reformulation of RNNs and CNNs as selective state-space models. The novelties brought by this architecture consist in:

- *A selection mechanism:* the selection mechanism in the Mamba model enables it to dynamically focus on relevant information and ignore irrelevant data. This mechanism is akin to the attention mechanism used in Transformer models. It allows the model to adjust its focus based on the input data, enhancing its ability to capture important features and dependencies;
- *Hardware awareness:* the Mamba architecture is optimized for GPU usage, which allows it to scale linearly with sequence length. This optimization ensures that the model can handle long sequences efficiently and benefit from parallel computing, resulting in faster training.

The Mamba architecture was originally developed for use in large language models (LLMs). However, recent studies in the literature have demonstrated its capability to perform segmentation on 3D medical images, showcasing its versatility and robustness in handling complex data structures. Therefore, the present study aims at determining whether a Mamba-based model can be effectively applied to complex time-series datasets, such as anomalous Brownian trajectories. The approach we chose was to evaluate the performances of Bi-Mamba, i.e. the novel state-space deep-learning architecture (Mamba) articulated with a bidirectional scan mechanism. Indeed, to the best of our knowledge, the Mamba architecture has not been used so far in the case of physical studies, and AnDi data in particular.

2.1. The Bi-Mamba block

Originally, the Mamba architecture has been designed to process 1D data sequence, in one direction. Considering the complex patterns that can occur in anomalous Brownian trajectories, we have implemented a bidirectional version of the Mamba block. As described in the algorithm 1, the block is made as an encoder and takes as an input $x \in \mathbb{R}^{(B,L,D)}$ with B the number of trajectories to be analyzed in parallel, L the length of the trajectories and D the number of features extracted from the trajectories. The output of the Bi-Mamba block is $y \in \mathbb{R}^{(B,L,D)}$. The Bi-Mamba block is composed of two Mamba blocks, one for the forward pass and the other for the backward pass. The forward Mamba block processes the input data in the forward direction, while the backward Mamba block processes the input data in the reverse direction. We denote the time-reversal operation as Flip, and the variables of each side as x_{forward} and x_{backward} , respectively. The outputs of the two Mamba blocks are then added and passed to a feedforward layer for further processing. To reduce the risk of vanishing gradients, residual connection between the input and the output of each step is applied [32].

**Algorithm 2.** Our Model—MambaBrown.

Input: $x \in \mathbb{R}^{(B,L,D)}$
Output: (y_S, y_K, y_α)

- 1 $y_S \leftarrow \text{BiMambaBlock}(x)$;
- 2 $y_S \leftarrow \text{Dense}(y_S)$; // $(B, L, 2)$
- 3 $y_K \leftarrow \text{BiMambaBlock}(x, y_S)$;
- 4 $y_K \leftarrow \text{Dense}(y_K)$; // $(B, 2)$
- 5 $y_K \leftarrow \text{ReLU}(y_K)$;
- 6 $y_\alpha \leftarrow \text{BiMambaBlock}(x, y_S)$;
- 7 $y_\alpha \leftarrow \text{Dense}(y_\alpha)$; // $(B, 2)$
- 8 $y_\alpha \leftarrow \text{Sigmoid}(y_\alpha)$;
- 9 **return** (y_S, y_K, y_α) ;

2.2. Our model—MambaBrown

To achieve the different tasks of the AnDi-2 challenge, we have implemented a model using the Bi-Mamba block, as shown in figure 2. The input data is first processed by a Bi-Mamba block for the segmentation of the diffusion state of the trajectory. As the trajectories are simulated using two different diffusion states, the segmentation block output is then processed by a dense layer to reduce the output dimensionality to $(B, L, 2)$, with each dimension corresponding to the probability of the trajectory to be in one of the two states. Then, the original features and the segmentation result are passed into two separate blocks, one for the K regression and the other for the α regression. Therefore, each task is handled by a dedicated block, allowing each block to specialize in a single function. Each of the outputs of the regression blocks is then processed by a dense layer to reduce the dimensionality of the output to $(B, 2)$, with each dimension corresponding to the predicted value of K and α respectively. The output of the α -regression block is passed to a sigmoid activation function and multiplied by 2 to ensure that the output is between 0 and 2. The output for the K -regression block is passed to a ReLU activation function to ensure that the output is positive.

3. Training and results

The data ensemble consists of sets of two-dimensional trajectories with coordinates $r(t) = [r_x(t), r_y(t)]$, with a maximum length of 200 time steps. The features computed using given trajectories are: the displacements for the first lag time $\Delta t = 1$, $\Delta r_i(\Delta t = 1) = r_i(t + \Delta t) - r_i(t)$, with $i \in \{x, y\}$, the one-dimensional MSDs $\langle \Delta r_i^2 \rangle(\Delta t)$, the displacement angle between two consecutive displacements [33], and the total displacement from the origin $d = \sqrt{x(t)^2 + y(t)^2}$. Trajectories smaller than 200 data points are zero-padded. The loss is then calculated by summing the losses of the three blocks and back-propagated using the Adam optimizer. The α loss is evaluated using mean absolute error (MAE), the K loss is evaluated using mean-squared

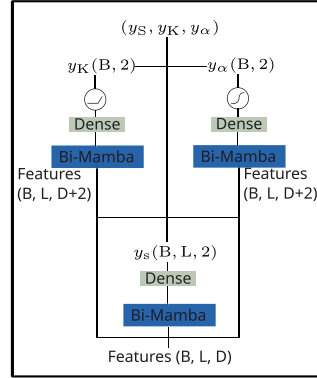


Figure 3. MambaBrown model implementation. Dense corresponds to the dense layer reducing the dimensionality of the Bi-Mamba block output. Circles correspond to the activation functions ReLu and Sigmoid, on the left and right respectively.

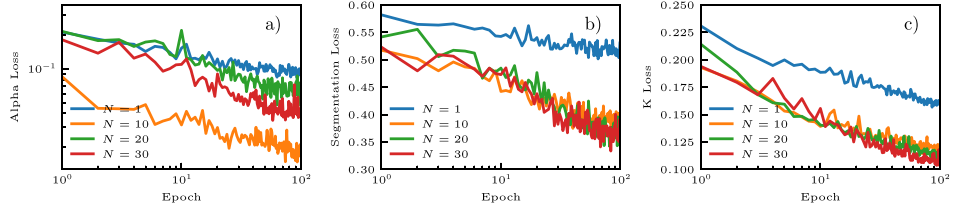


Figure 4. Loss values calculated on a test dataset consisting of 200 000 trajectories, as functions of the number of epochs for different depths. Specifically, we show: (a) the Mean Absolute Error (MAE) loss for α inference, (b) the Mean-Squared Logarithmic Error (MSLE) loss for K inference, and (c) the Weighted Cross-Entropy (WCE) loss for diffusion-state segmentation.

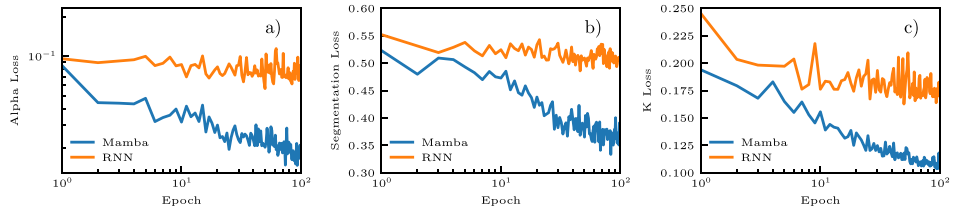


Figure 5. Loss values calculated on a test dataset consisting of 200 000 trajectories, as functions of the number of epochs for our Mamba-based and RNN-based models. Specifically, we show: (a) the Mean Absolute Error (MAE) loss for α inference, (b) the mean-squared logarithmic error (MSLE) loss for K inference, and (c) the weighted cross-entropy (WCE) loss for diffusion-state segmentation.

logarithmic error (MSLE), and the segmentation loss is evaluated using the weighted cross-entropy error (WCE). The training is carried out on a subset of 200 000 trajectories generated using the AnDi-2 challenge simulation tools. The loss is calculated by summing the losses of the three blocks and back-propagated using the Adam optimizer. Segmentation is evaluated using the WCE loss, while the K regression is evaluated using the MSLE loss, and the α regression is evaluated using the MAE loss. In figure 4, we evaluate the loss as a function of the depth N of the different blocks, from which we determine our best model to have a depth of $N = 10$ for α regression and $N = 30$ for segmentation and K regression.

For fair benchmarking purposes, we compare our model (see figure 3) to a similar architecture but using bidirectional RNN instead of our Bi-Mamba block (see figure 2). As shown in figure 5, the Bi-Mamba model shows promising performances, scoring better overall in each category than a RNN-based model. Moreover, it shows its capabilities to train for a larger number of epochs without overfitting and with a smaller loss variance.

4. Conclusion

In this work, we implemented the structure and evaluated the performances of the Mamba architecture—a novel application of state-space models—towards the characterization of AnDi. Using a bidirectional scan mechanism, we demonstrated a notable efficiency in diffusion-state segmentation, as well as for effective-diffusion-coefficient and anomalous-exponent regression tasks. Improvement could be achieved through the use of specialized models, fine-tuned using simulations generated near the predictions of the generally-trained model presented in this work. Our contribution to the AnDi-2 challenge was designed in order to specifically evaluate the performances of Bi-Mamba models for AnDi. Our first model developed during the competition phase ranked 7th in α inference (MAE of 0.27), 9th in K inference (MSLE of 0.05), 3rd in diffusion-type measurement [i.e. trapped with ($\alpha < 0.2$), directed with $\alpha > 1.8$, or normal with $\alpha = 1$ and with a F1 error of 0.91], and 10th on the change-point detection (root-mean-squared error of 2.7). Moreover, we expect improvements on the segmentation task by using Mamba-based Unet methods [34]. All together, this work contributes to the knowledge on, and continued development of, deep-learning methods for AnDi and physics in general.

Data availability statement

The data that support the findings of this study are openly available at the following URL: <https://github.com/EMetBrown-Lab/Mamba-EMetBrown-ANDI2>.

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ORCID iDs

Maxime Lavaud  <https://orcid.org/0000-0002-1463-2135>

Yosef Shokeeb  <https://orcid.org/0009-0003-6040-4103>

Juliette Lacherez  <https://orcid.org/0009-0002-9580-5735>

Yacine Amarouchene  <https://orcid.org/0000-0002-0475-5510>

Thomas Salez  <https://orcid.org/0000-0001-6111-8721>

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Enhanced diffusion over a periodic trap by hydrodynamic coupling to an elastic mode



Juliette Lacherez¹, Maxime Lavaud^{1,2}, Yacine Amarouchene¹✉, David S. Dean¹✉ & Thomas Salez¹✉

In many physical systems, degrees of freedom are coupled *via* hydrodynamic forces, even in the absence of Hamiltonian interactions. A particularly important and widespread example concerns the transport of microscopic particles in fluids near deformable boundaries. In such a situation, the influence of elastohydrodynamic couplings on Brownian motion remains to be understood. Unfortunately, the temporal and spatial scales associated with the thermal fluctuations of usual surfaces are often so small that their deformations are difficult to monitor experimentally, together with the much slower and larger particle motion at stake. Here, we propose a minimal model describing the hydrodynamic coupling of a colloidal particle to a fluctuating elastic mode, in presence of an external periodic potential. We demonstrate that the late-time diffusion coefficient of the particle increases with the compliance of the elastic mode. Our results reveal and quantify two features: first, spontaneous microscopic transport in complex environments can be affected by soft boundaries – a situation with numerous practical implications in nanoscale and biological physics; and second, the effects of fast and tiny surface deformations are imprinted in long-term and large-distance colloidal mobility, and are therefore measurable in practice.

Brownian motion refers to the random movement of a colloid at equilibrium, induced by its collisions with the molecules of the surrounding fluid. It has played a foundational role in modern statistical physics since its discovery¹ and theoretical description^{2,3} over a century ago. The celebrated experiments of Perrin on colloids in a gravitational field⁴ validated further the theoretical description and confirmed the atomic nature of matter. Extending the classical framework of Brownian motion in bulk Newtonian fluids to more realistic systems, such as particles diffusing in complex environments, opens up new challenges for nanoscale and biological physics. One common route is to study Brownian motion in viscoelastic fluids or intracellular media, as typically performed through microrheology⁵, which revealed the emergence of anomalous diffusion⁶. Another source of complexity lies in the introduction of topological constraints and boundaries. For instance, a simple-yet-canonical situation where interfaces matter is the diffusion near rigid walls, that has been extensively studied both theoretically and experimentally^{7–15}. Besides the practical interest of such a configuration towards the sensing of surface forces¹⁶, one of the key aspects lies perhaps in the apparition of anisotropic and space-dependent mobilities¹⁷ due to the frictional boundary condition at the walls. As a consequence, multiplicative noises emerge, leading to non-Gaussian diffusion and the probability enhancement of rare events^{18–22}.

However, in most practical situations where molecules or micro-organisms diffuse near surfaces, the latter are typically soft. Therefore, the dynamical properties of Brownian particles near deformable boundaries must be understood. Along this line of thought, a few studies, mostly theoretical, have addressed the case of tracers diffusing near soft surfaces. Examples include liquid interfaces^{23,24}, and fluctuating ones²⁵. The case of elastic interfaces was also addressed in the contexts of contactless microrheology²⁶ and biological membranes²⁷. Nevertheless, these works focused on the low-coupling regime, where the particles are distant or point-like, and thus unable to generate rectified flows that lead to surface deformation. Interestingly, though, within a purely deterministic context, near-contact elastohydrodynamic couplings have been studied in the past decades. Examples include contactless colloidal-probe rheology of soft substrates in normal mode^{28,29}, as well as the emergence of lift forces at low Reynolds numbers^{30,31}, that were first revealed theoretically^{32–36}, and then confirmed experimentally^{37–41}. The leading-order corrections induced by elasticity to the full mobility tensor near a rigid wall⁷ were obtained for various geometries and rheologies^{42–44}, and revealed a zoology of counter-intuitive inertial-like solutions.

In view of the above state of the art, the arising challenge is now to understand the influence of soft boundaries on the spontaneous motion of

¹Univ. Bordeaux, CNRS, LOMA, Talence, France. ²Univ. Bordeaux, CNRS, Bordeaux INP, CBMN, Pessac, France. ✉e-mail: yacine.amarouchene@u-bordeaux.fr; david.dean@u-bordeaux.fr; thomas.salez@cnrs.fr

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nearby finite-sized Brownian particles. Molecular-dynamics simulations were performed in such a context⁴⁵. Besides, a recent experimental study indicated that soft colloids diffusing near rigid walls feel transient, non-conservative forces that could be related to an intricate combination of thermal fluctuations, soft boundaries and viscous flows⁴⁶. Finally, at a practical level, a natural question to ask is whether the effects of fast and tiny surface fluctuations can be observed over the long times and large distances associated with colloidal diffusion.

Here, we introduce a minimal, but generic, theoretical model in order to investigate Brownian elastohydrodynamic effects. Specifically, we combine the hydrodynamic coupling of a colloidal particle with an elastic mode^{47,48} to an external periodic trapping^{49–52}. We solve the problem both analytically and numerically, and demonstrate that the late-time diffusion coefficient of the particle increases notably with the compliance of the elastic mode.

Results

Model

As depicted in Fig. 1, we consider an overdamped system made of two coupled degrees of freedom: i) a principal degree of freedom $q_1(t)$, corresponding to the spatial position of a test colloid over time t , which is free to explore space while being subject to a periodic trapping potential $\phi(q_1)$ with spatial period λ (in practice, such trapping potential could represent a surface with patchy, adhesive patterns); and ii) an auxiliary degree of freedom $q_2(t)$, corresponding to an elastic mode, which is harmonically confined with a potential energy $q_2^2/(2\kappa)$, where κ is the compliance. The compliance, i.e., inverse stiffness, represents the softness parameter of the elastic mode.

Thus, the total energy of the system is given by the Hamiltonian

$$H(q_1, q_2) = \phi(q_1) + \frac{q_2^2}{2\kappa}. \quad (1)$$

At thermal equilibrium, the probability density function (PDF) of q_1 and q_2 is given by the Gibbs-Boltzmann distribution

$$P_{\text{eq}}(q_1, q_2) = \frac{\exp[-\beta H(q_1, q_2)]}{Z}, \quad (2)$$

where Z is the normalizing partition function, and where $\beta = 1/(k_B T)$ with T the temperature and k_B Boltzmann's constant. The equilibrium statistics of q_1 and q_2 are independent due to the additivity of the Hamiltonian, which implies, in particular, that q_2 does not affect the equilibrium distribution of q_1 .

The system dynamics is described by the coupled overdamped Langevin equations

$$\gamma_{11} \frac{dq_1}{dt} + \gamma_{12} \frac{dq_2}{dt} = -\phi'(q_1) + \eta_1(t), \quad (3)$$

$$\gamma_{12} \frac{dq_1}{dt} + \gamma_{22} \frac{dq_2}{dt} = -\frac{q_2}{\kappa} + \eta_2(t), \quad (4)$$

where the constants γ_{ij} , with $i, j \in \{1, 2\}$, are the components of the friction tensor γ , which is related to the mobility tensor μ through $\mu = \gamma^{-1}$; and where the functions $\eta_i(t)$ are zero-mean Gaussian white noises, with a correlation function $\langle \eta_i(t) \eta_j(t') \rangle = 2k_B T \delta(t - t') \gamma_{ij}$, where $\langle \cdot \rangle$ is the ensemble average. This choice of noise correlation is self-consistent with the equilibrium distribution of q_1 and q_2 being given by Eq. (2).

Theoretical analysis

Let us now define

$$\epsilon = \frac{\kappa k_B T}{\lambda^2} \quad (5)$$

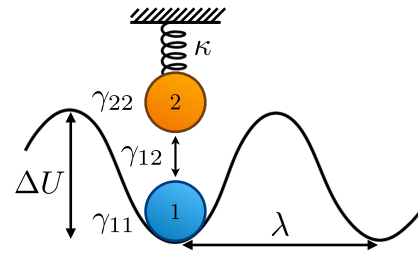


Fig. 1 | Schematic of the system. A test Brownian particle (1) is trapped in a periodic energy potential of amplitude ΔU and spatial period λ . It is coupled via a hydrodynamic-like interaction of constant damping coefficient γ_{12} to a thermally-fluctuating elastic mode, represented by another Brownian particle (2) trapped in a harmonic trap of compliance (i.e., inverse softness) κ . Besides, both particles feel individual hydrodynamic drag forces represented by their constant self-drag coefficients γ_{11} and γ_{22} , and are described within the overdamped Langevin regime.

the dimensionless compliance. In the weakly-compliant regime ($\epsilon \ll 1$), one can compute the late-time diffusion coefficient $D^*(\epsilon)$ of the colloid using a perturbation analysis at $\mathcal{O}(\epsilon)$, together with an exact representation of the effective diffusion coefficient obtained via the Stokes-Einstein relation, and a multiscale analysis^{53,54}. Doing so (see details in SI Note 3), we obtain

$$D^*(\epsilon) \simeq D^*(0)(1 + \alpha), \quad (6)$$

with

$$\alpha = \beta \epsilon \frac{\lambda^2 \gamma_{12}^2 \int_0^\lambda dq_1 \phi'(q_1)^2 \exp[\beta \phi(q_1)]}{\gamma_{11}^2 \int_0^\lambda dq_1 \exp[\beta \phi(q_1)]}, \quad (7)$$

and

$$D^*(0) = \frac{k_B T \lambda^2}{\gamma_{11} Z_+ Z_-}, \quad (8)$$

where

$$Z_\pm = \int_0^\lambda dq_1 \exp[\pm \beta \phi(q_1)]. \quad (9)$$

Since $\alpha \geq 0$, the hydrodynamic coupling of the colloid to the elastic mode always increases the late-time diffusion coefficient of the colloid. Moreover, the effect is larger at larger compliance κ . We also see that a non-flat periodic potential $\phi(q_1)$ is required for the effect to exist, within the assumptions of the model. Such an influence of external potentials on hydrodynamic friction is reminiscent of a molecular-dynamics study in the literature⁵⁵.

In the particular case of a sinusoidal potential, where $\phi(q_1) = \Delta U \sin(2\pi q_1/\lambda)$, with ΔU the amplitude of the potential, we have

$$\alpha = 4\pi^2 \epsilon \frac{\gamma_{12}^2 v I_1(v)}{\gamma_{11}^2 I_0(v)}. \quad (10)$$

with $v = \beta \Delta U$, where I_n denotes the order- n modified Bessel function of the first kind.

As such, the relative increase of the late-time diffusion coefficient is larger for smaller λ . This is attributed to the fact that for a smaller λ the coupling to the elastic mode increases the jump rate of the colloid between neighboring minima, instead of being averaged out as would be the case at large λ . We also see that the relative increase of the late-time diffusion coefficient depends linearly on the function $v I_1(v)/I_0(v)$, which is a monotonically increasing function of v . This means that the relative magnitude of the effect increases with the depth of the trap – the obvious counterpart

being that larger times are then needed to reach the late-time diffusive regime. Finally, the relative magnitude of the effect increases with the relative magnitude γ_{12}/γ_{11} of the hydrodynamic coupling to the elastic mode, as expected.

The multiscale analysis can also be employed in the limit where $\epsilon \rightarrow \infty$, which corresponds to a very compliant elastic mode. In this case, we also find (see details in SI Note 4) that the late-time diffusion coefficient is increased relatively to the rigid-case value, and saturates to a maximum value at large ϵ , as

$$D^* = \frac{k_B T \mu_{11} (\mu_{11} \mu_{22} - \mu_{12}^2)}{\lambda^{-2} Z_+ Z_- (\mu_{11} \mu_{22} - \mu_{12}^2) + \mu_{12}^2} + O\left(\frac{1}{\sqrt{\epsilon}}\right). \quad (11)$$

Once again, we first note that when the periodic potential $\phi(q_1)$ is flat, the late-time diffusion coefficient is not modified by the hydrodynamic coupling to the elastic mode. However, interestingly, such a plateau value can be notably higher than the rigid-case value, indicating the potential strength of the effect in practical systems relevant to soft matter and biophysics. For instance, let us consider a spherical particle of radius $a = 100$ nm within a fluid of viscosity $\eta = 10^{-3}$ Pa.s and at temperature $T = 300$ K. The particle is assumed to be at a distance $d = 10$ nm from a soft tissue with a Young modulus $E = 1$ kPa, or equivalent stiffness $\kappa^{-1} \sim E\sqrt{ad}$, and is placed in a trapping potential of amplitude $4k_B T$ with spatial period $\lambda = 100$ nm. Taking $\mu_{11} = \mu_{22} = (6\pi\eta a)^{-1}$ for the self-mobilities, and Oseen's $\sim \mu_{12} = (4\pi\eta r)^{-1}$ for the coupling mobility, with $r = 2a + d$, one expects a 100% relative increase of the long-term diffusivity with respect to the classical Lifson-Jackson value.

Heuristic argument

In order to understand the softness-induced late-time diffusivity enhancement from a physical point of view, we now present a much simpler heuristic argument giving the same result as the rigorous analysis in the small- ϵ regime (see details in SI Note 2). The intrinsic time scale $\tau_2 = \kappa/\mu_{22}$ of the stochastic process $q_2(t)$ is assumed to be small with respect to all the other time scales. In this limit, the effective zero-temperature overdamped Langevin equation for $q_1(t)$ is

$$\frac{dq_1}{dt} = -\mu_e(q_1)\phi'(q_1), \quad (12)$$

where we have introduced the dressed mobility

$$\mu_e(q_1) = \frac{1}{\gamma_{11} \left[1 + \frac{\kappa \gamma_{12}^2 \phi''(q_1)}{\gamma_{11}} \right]}. \quad (13)$$

The equilibrium marginal PDF in position q_1 is $P_{\text{meq}}(q_1) = (Z_-)^{-1} \exp[-\beta\phi(q_1)]$. Therefore, assuming that the time-dependent marginal PDF $P_1(q_1, t)$ satisfies an effective Fokker-Planck equation, the latter must have

the form

$$\frac{\partial P_1(q_1, t)}{\partial t} = \frac{\partial}{\partial q_1} \left\{ \mu_e(q_1) \left[k_B T \frac{\partial P_1(q_1, t)}{\partial q_1} + \phi'(q_1) P_1(q_1, t) \right] \right\}, \quad (14)$$

where the T -independent term on the right-hand side is determined from the zero-temperature dynamics, while the T -dependent term on the right-hand side is constructed to ensure convergence of P_1 to P_{meq} at infinite time. Note that we have implicitly assumed here that the dressed mobility $\mu_e(q_1)$ does not exhibit any temperature dependence.

The dressed diffusion coefficient is given by the Stokes-Einstein relation, $D_e(q_1) = k_B T \mu_e(q_1)$, which differs from the bare (or “microscopic”) diffusion coefficient $D_m = k_B T \mu_{11}$ of the colloid. Interestingly, the spatial dependence of the dressed diffusion coefficient implies that the effective Langevin equation must have a multiplicative noise, leading to a supplementary source of non-Gaussian displacements^{16,18–22,56} in addition to the trivial one induced by the trapping potential.

We now invoke the Lifson-Jackson formula⁵⁷, providing the late-time diffusion coefficient in a periodic potential, which reads

$$D^*(\epsilon) = \frac{\lambda^2}{\int_0^\lambda dq_1 \frac{\exp[\beta\phi(q_1)]}{D_e(q_1)} \int_0^\lambda dq_1 \exp[-\beta\phi(q_1)]}. \quad (15)$$

From this expression, we can recover Eqs. (6) and (7). We emphasize that the heuristic argument we presented here is not rigorous. Nevertheless, it can also be applied to evaluate the effective diffusion coefficient of an underdamped Brownian particle near the overdamped limit (see details in SI Note 6), for which we once again recover the results obtained via rigorous multiscale analysis^{53,54}. The same method also yields the first-order correction to the effective diffusion coefficient for a particle carrying a dipole moment which interacts with a random or periodic electric field when the dipole dynamics is assumed to be much faster than the particle diffusion⁵⁸.

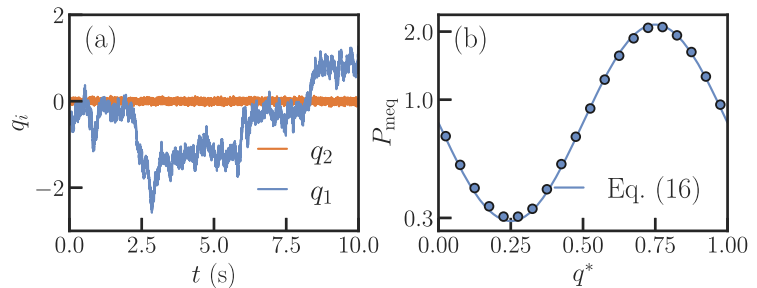
Comparison to numerical simulations

In order to verify the above predictions, Eqs. (3) and (4) are numerically solved for a sinusoidal trapping potential $\phi(q_1)$ (see details in Methods). Typical raw trajectories are shown in Fig. 2a). We observe that the elastic mode $q_2(t)$ fluctuates around its equilibrium value ($q_2 = 0$), within the harmonic potential it is subject to, whereas the position $q_1(t)$ of the test colloid fluctuates at short times in a minimum of the sinusoidal potential $\phi(q_1)$ before randomly jumping from one minimum to the other by passing the potential barrier. From such trajectories, we can first compute the late-time PDF in positions. As an example, Fig. 2b) shows the equilibrium marginal PDF of the periodized variable q^* , defined by $q_1 = q^* \bmod(\lambda)$, which satisfies

$$P_{\text{meq}}(q^*) = \frac{\exp[-\beta\phi(q^*)]}{\int_0^\lambda dq \exp[-\beta\phi(q)]}. \quad (16)$$

Fig. 2 | Trajectories and equilibrium distribution.

a Typical trajectories $q_i(t)$ of the test particle ($i = 1$) and the elastic mode ($i = 2$), simulated by numerically solving Eqs. (3) and (4), for mobility matrix elements $\mu_{11} = 1.0$, $\mu_{12} = 0.5$, $\mu_{22} = 0.5$, with $\Delta U = 1.0$, $\lambda = 1.0$, $\kappa = 1.0 \times 10^{-3}$, time increment $\Delta t = 1.0 \times 10^{-4}$, and $k_B T = 1.0$. Simulations are initialized from equilibrium. **b** Late-time marginal PDF P_{meq} for the periodized position q^* , given by $q_1 = q^* \bmod(\lambda)$, of the test particle, obtained by binning 10^5 trajectories of 10^7 points. The symbols are the results of the numerical simulations. The solid blue line corresponds to the Boltzmann distribution given in Eq. (16).



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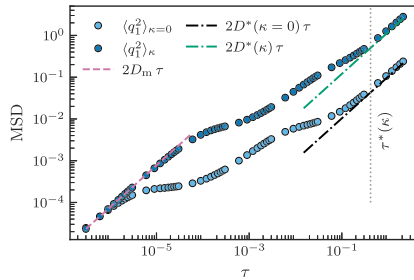


Fig. 3 | Effect of softness on the mean-squared displacement. Mean-squared displacements $\langle q_1^2 \rangle_\kappa$ and $\langle q_1^2 \rangle_0$ of the test particle as functions of the time increment τ , for $\kappa = 0.1$ (dark blue disks) and $\kappa = 0.0$ (light blue disks) respectively, for damping matrix elements $\gamma_{11} = 1.309$, $g_{12} = 1.296$, $\gamma_{22} = 1.309$, with $\Delta U = 2.7$, $\lambda = 0.5$, time increment $\Delta t = 3.0 \times 10^{-7}$, and $k_B T = 1.0$. The symbols are the results of the numerical simulations. The dash-dotted black line indicates $2D^*(0)\tau$, where $D^*(0)$ is the classical Lifson-Jackson late-time diffusion coefficient⁵⁷. The dash-dotted green line indicates $2D^*(\kappa)\tau$, where $D^*(\kappa)$ is the late-time diffusion coefficient given by Eq. (6). The dashed pink line indicates $2D_m\tau$, where $D_m = k_B T \mu_{11}$ is the bare diffusion coefficient. The characteristic time $\tau^*(\kappa) = \lambda^2/D^*(\kappa)$ is indicated with the light-gray vertical dotted line.

In order to probe the dynamical effects of the hydrodynamic coupling of the trapped test particle to the elastic mode, we now turn to the analysis of the mean-squared displacement (MSD), defined as

$$\langle q_i^2 \rangle(\tau) = \langle [q_i(t + \tau) - q_i(t)]^2 \rangle, \quad (17)$$

where $\langle \cdot \rangle$ denotes the ensemble average, and τ is a time increment. As shown in Fig. 3, the MSD of the test colloidal particle exhibits two distinct diffusive regimes separated by a crossover region.

We compute from the MSD the short-time diffusive regime, characterized by the bare diffusion coefficient D_m , and the late-time diffusive regime, which is characterized by the late-time diffusion coefficient D^* , and is reached when the particle has diffused over many periods of the trapping potential, i.e., when its MSD is large compared to λ^2 . This sets a time scale $\tau^*(\kappa) = \lambda^2/D^*(\kappa)$, which must be exceeded to see the late-time diffusive regime. We extract the relative excess in late-time diffusion coefficient, i.e., $\Delta D^*/D^* = [D^*(\epsilon) - D^*(0)]/D^*(0)$, as a function of ϵ , from log-log fits of the MSD, excluding: data below $5\tau^*$, which belongs to a different dynamical regime, as well as from the last decade, which lacks statistics for proper data processing. The results are shown in Fig. 4 for a given set of parameters. We observe that the numerical-simulation data agree well with the small- ϵ and large- ϵ asymptotic predictions of Eqs. (6) and (11), respectively. Furthermore, the Kubo formula can be used to evaluate D^* for any value of ϵ (see details in SI Note 5, and Eq. (S65)). We see that the numerical-simulation data agree at all ϵ values with the Kubo formula, and that the latter interpolates well the two asymptotic predictions.

Conclusions

We theoretically and numerically studied the Brownian motion of colloid-like particles near elastic interfaces using minimal ingredients. We showed that a Brownian particle trapped in a periodic potential sees its late-time diffusivity dressed and significantly enhanced by hydrodynamic coupling to a hidden elastic mode. Despite being minimalist, our description reveals generic, original and large effects, relevant for the transport of colloidal particles near complex boundaries. In particular, our results suggest the intriguing possibility of understanding, and even tuning, the diffusivity of colloids near deformable and patchy interfaces—with key implications for biophysics and nanofluidics.

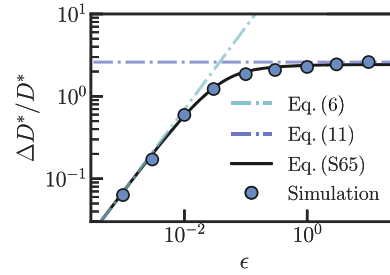


Fig. 4 | Effect of softness on diffusivity. Relative increase $\Delta D^*/D^* = [D^*(\epsilon) - D^*(0)]/D^*(0)$ of the late-time diffusion coefficient D^* of the test particle as a function of the dimensionless compliance ϵ , for $\mu_{11} = 1.0$, $\mu_{12} = 0.85$, $\mu_{22} = 1.0$, $\Delta U = 3.0$, $\lambda = 1.0$, and $k_B T = 1.0$. The symbols are the results of the numerical simulations. The dash-dotted sky blue line corresponds to the small- ϵ prediction of Eq. (6). The dash-dotted navy blue line corresponds to the large- ϵ prediction of Eq. (11). The solid black line corresponds to the Kubo formula of Eq. (S65) (see details in SI Note 5).

Methods

Equations (3) and (4) are numerically solved for a sinusoidal trapping potential $\phi(q_1)$, using an Euler scheme, through a compiled *Cython*⁵⁹ code. The discretized Langevin equations read

$$q_1(t + \Delta t) - q_1(t) = -\mu_{11}\phi'(q_1(t))\Delta t - \frac{\mu_{12}}{\kappa}q_2(t)\Delta t + \xi_1\sqrt{2k_B T\Delta t}, \quad (18)$$

$$q_2(t + \Delta t) - q_2(t) = -\frac{\mu_{22}}{\kappa}q_2(t)\Delta t - \mu_{12}\phi'(q_1(t))\Delta t + \xi_2\sqrt{2k_B T\Delta t}, \quad (19)$$

where ξ_i represent zero-mean Gaussian random variables, characterized by the correlator

$$\langle \xi_i \xi_j \rangle = \mu_{ij}. \quad (20)$$

We carefully choose the time step Δt for numerical stability and accuracy, satisfying $\Delta t \leq 10^{-2}\tau_2$. The mobility tensor μ_{ij} must be positive-definite. To enforce this, we write noises ξ_i as $\xi_i = m_{ij}\eta_j$, with η_j zero-mean, independent Gaussian random variables satisfying $\langle \eta_i \eta_j \rangle = \delta_{ij}$, with $\delta_{ij} = 1$ for $i = j$ and 0 otherwise. This yields a decomposition of the mobility tensor $\mu_{ij} = m_{ik}m_{kj}$. The matrix elements m_{ij} ensure that noise correlations are respected. For computational efficiency, the noise array is pre-generated using the `numpy.random.normal` function. The simulation recursively updates the system state at each discrete time step. Initial positions of both modes are sampled from Eq. (2) to ensure thermalization. Trajectories are computed iteratively, and their corresponding displacement PDFs are binned. Convergence is evaluated with the relative distance between iterations of the stacked displacement PDFs, with a criterion of 1% relative distance between the stacked histogram and new iterations, for more than three consecutive iterations.

Data availability

All data are available upon reasonable request to the corresponding authors. A Supplementary Information (SI) document provides details on: the analytical derivation, and numerical estimations.

Code availability

The simulation scripts are open-source and available on the GitHub repository https://github.com/EMetBrown-Lab/Lacherez2024_Codes. The repository provides a *Jupyter* notebook for single-trajectory simulations. For large-scale parameter sweeps, the `setup.py` script enables optimized

pre-compilation using *Cython*, and the `automate_simulation.py` script supports systematic exploration of the parameter space. These tools are designed for efficient single-node execution and ensure reproducibility of all results presented in this study. We invite the reader to consult the README.md file. Additional treatment codes are available upon reasonable request to the authors.

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Author contributions

J.L. conducted numerical simulations, data analysis and drafted the manuscript. M.L. helped with the optimization of the numerical simulations and commented on the manuscript. Y.A. conceived the project, numerically computed the Kubo formula and commented on the manuscript. D.S.D. conceived the project, developed the analytical model and edited the manuscript. T.S. conceived the project, led the project, and edited the manuscript.

Competing interests

The authors declare no competing interests.

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Correspondence and requests for materials should be addressed to Yacine Amarouchene, David S. Dean or Thomas Salez.

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Enhanced diffusion over a periodic trap by hydrodynamic coupling to an elastic mode
– Supplementary Information –

Juliette Lacherez¹, Maxime Lavaud^{1,2}, Yacine Amarouchene¹✉, David S. Dean¹† and Thomas Salez¹‡

¹Univ. Bordeaux, CNRS, LOMA, UMR 5798, F-33400, Talence, France and
²Univ. Bordeaux, CNRS, Bordeaux INP, CBMN, UMR 5248, F-33600, Pessac, France
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I. SUPPLEMENTARY NOTE 1: DESCRIPTION OF THE MODEL

We consider an overdamped system made up of two degrees of freedom, the principal degree of freedom q_1 which is free to explore a large region of space, for example subjected to a periodic potential $\phi(q_1)$, and the second degree of freedom q_2 , highly constrained and taken to be harmonically confined with potential energy $\frac{1}{2\kappa}q_2^2$, where we assume that κ the spring compliance is small. The total energy of the system is thus given by the Hamiltonian

$$H(q_1, q_2) = \phi(q_1) + \frac{1}{2\kappa}q_2^2. \quad (\text{S1})$$

In thermal equilibrium, the steady state of the probability distribution of q_1 and q_2 is given by the Gibbs-Boltzmann distribution

$$P_{\text{eq}}(q_1, q_2) = \frac{\exp[-\beta H(q_1, q_2)]}{Z}, \quad (\text{S2})$$

where Z is the normalizing partition function and $\beta = 1/(k_B T)$ is the inverse thermal energy, where T is the temperature and k_B is Boltzmann's constant. In equilibrium, the statistics of q_1 and q_2 are independent due to the additivity of the Hamiltonian H , *i.e.* $P_{\text{eq}}(q_1, q_2) = P_{1\text{eq}}(q_1)P_{2\text{eq}}(q_2)$. However, dynamically, they can be coupled; for instance this model can be seen as a toy model for a colloidal particle near an elastic compliant surface. As such, any single time observation in equilibrium of the measurable quantity q_1 will be independent of the effect of q_2 and hence will not show any effect due to the hydrodynamic coupling with q_2 . This means that one would have to measure at least two time correlation functions; but, due to the rapid dynamics of q_2 , one would have to resolve the dynamics on very short time scales to detect the hydrodynamic coupling.

The simplest form for the over-damped Langevin dynamics of this system is

$$\frac{dq_1}{dt} = -\mu_{11}\phi'(q_1) - \frac{\mu_{12}}{\kappa}q_2 + \eta_1(t) \quad (\text{S3})$$

$$\frac{dq_2}{dt} = -\frac{\mu_{22}}{\kappa}q_2 - \mu_{12}\phi'(q_1) + \eta_2(t), \quad (\text{S4})$$

where the mobility coefficients μ_{ij} are taken for simplicity to be constant, and the noise correlation function is chosen to ensure that this Langevin system has a steady state distribution given by Eq. (S2), explicitly

$$\langle \eta_i(t)\eta_j(t') \rangle = 2k_B T \delta(t - t')\mu_{ij}. \quad (\text{S5})$$

From a mathematical standpoint, the constraints on the mobility matrix are that it is symmetric and has positive eigenvalues. However, from a physical point of view, in the absence of coupling, we must have

$$\mu_{11} > 0, \text{ and } \mu_{22} > 0. \quad (\text{S6})$$

The constraints on the eigenvalues of the mobility matrix then translate to

$$\det(\mu) = \mu_{11}\mu_{22} - \mu_{12}^2 > 0. \quad (\text{S7})$$

In an experimental context, it may be possible to directly observe the motion of the variable q_1 , but not necessarily on very short time scales, while the direct observation of the second variable q_2 may be impossible for it cannot be

* yacine.amarouchene@u-bordeaux.fr

† david.dean@u-bordeaux.fr

‡ thomas.salez@cnrs.fr

resolved spatially nor temporally. If the variable q_1 is subject to a confining potential and a measurement is made of its equilibrium statistics, the variable q_1 has no effect whatsoever, and changing the value of κ for instance will not affect the measurements. Observing the dynamics of the variable q_1 may yield information about the variable q_2 but if the time scale over which q_2 equilibrates is too short, again this information is censored by being averaged out.

Here, we propose a solution to demonstrate the effect of the variable q_2 by examining its effect on the late time diffusion constant of q_1 when it is subjected to a periodic potential. Given that we are examining diffusion at large times and over large distances, neither the temporal nor the spatial resolutions of the experiment are an issue. The effect of the variable q_2 is determined by its influence on how q_1 moves in the potential ϕ and how this influence integrates over time to change the diffusion constant of q_1 . We will see that there is an intrinsic time scale associated with the dynamics of the elastic variable q_2 , and we assume that this time scale is small and thus difficult to resolve. Measurements of q_1 made over time scales larger than τ_2 will not see the effect of q_2 due to averaging. If the variable q_1 diffuses in a periodic potential and the typical time to cross the potential barriers (*i.e.* jump from minimum to minimum) is of the order of τ_2 , then the first passage time from minimum to minimum will be affected by the dynamics of q_2 and thus show up in the late time diffusion constant of q_1 , yielding a mechanism for revealing the hydrodynamic coupling. Rather than trying to resolve short time scales directly, we are thus imposing a short distance, and this can be done with an optical trap.

II. SUPPLEMENTARY NOTE 2: HEURISTIC ARGUMENT FOR THE CORRECTION TO THE LATE TIME DIFFUSION CONSTANT

To determine the effective Langevin dynamics for the variable q_1 in the limit where κ is small, and so the variable q_2 is small, we consider the dynamics of the system without any noise. Using the subsequently derived effective dynamics for q_1 at zero temperature, we then add a white noise to the system in order to ensure that the resulting Langevin equation has the correct marginal equilibrium probability distribution

$$P_{\text{meq}}(q_1) = \frac{\exp[-\beta\phi(q_1)]}{Z_-}, \quad (\text{S8})$$

where $\beta = \frac{1}{k_B T}$ is the inverse thermal energy, and where we use the notation (made dimensionless as compared to the main text)

$$Z_{\pm} = \int_0^{\lambda} dq \exp[\pm\beta\phi(q)], \quad (\text{S9})$$

where λ is the period of the potential. This is clearly not a rigorous procedure, and we will justify the result rigorously below using multiscale analysis. This method can also be applied to compute the correction to diffusion of a Brownian particle in a periodic potential due to inertia (the deviation from the strongly over-damped limit) [11, 12] and actually recovers first order correction (for completeness, we give a derivation later in this SM). It can also be applied to compute the correction to the diffusion constant of a Brownian particle carrying an electric dipole in a periodic electric field; and again, the simple analysis agrees with a more rigorous one using stochastic analysis [3].

We proceed by formally solving Eq. (S4) at $T = 0$ (that is to say, without noise) to give

$$q_2 = -\frac{\kappa\mu_{12}}{\mu_{22}} \left(1 + \frac{\kappa}{\mu_{22}} \frac{d}{dt}\right)^{-1} \phi'(q_1), \quad (\text{S10})$$

where $\tau_2 = \frac{\kappa}{\mu_{22}}$ is a short time scale due to the fact that κ is small. Carrying out an operator expansion for small τ_2 , we find

$$q_2 \simeq -\frac{\kappa\mu_{12}}{\mu_{22}} \left[\phi'(q_1) - \frac{\kappa}{\mu_{22}} \phi''(q_1) \frac{dq_1}{dt} + O(\tau_2^2) \right]. \quad (\text{S11})$$

Now, substituting this in Eq. (S3) gives

$$\frac{dq_1}{dt} = -\left(\mu_{11} - \frac{\mu_{12}^2}{\mu_{22}}\right) \phi'(q_1(t)) - \frac{\kappa\mu_{12}^2}{\mu_{22}^2} \phi''(q_1(t)) \frac{dq_1}{dt}, \quad (\text{S12})$$

and from this

$$\frac{dq_1}{dt} = -\mu_e(q_1) \phi'(q_1) \quad (\text{S13})$$

where $\mu_e(q_1)$ is the effective mobility

$$\mu_e(q_1) = \frac{\mu_{11}\mu_{22} - \mu_{12}^2}{\mu_{22} \left(1 + \frac{\kappa\phi''(q_1)\mu_{12}^2}{\mu_{22}^2} \right)} \quad (\text{S14})$$

In terms of the friction matrix $\gamma = \mu^{-1}$, the effective mobility can be written in a simpler form

$$\mu_e(q_1) = \frac{1}{\gamma_{11} \left(1 + \frac{\kappa\phi''(q_1)\gamma_{12}^2}{\gamma_{11}^2} \right)}. \quad (\text{S15})$$

Rather than discussing the noise necessary to render this into a Langevin equation, we note that to have the correct equilibrium distribution, the corresponding Fokker-Planck equation must read

$$\frac{\partial P_1(q_1, t)}{\partial t} = \frac{\partial}{\partial q_1} \mu_e(q_1) \left[k_B T \frac{\partial P_1(q_1, t)}{\partial q_1} + \phi'(q_1) P_1(q_1, t) \right], \quad (\text{S16})$$

and we see that the effective diffusion constant is given by $D_e(q_1) = k_B T \mu(q_1)$. Note that in doing this, we have added multiplicative noise and a temperature dependent drift if we use the Ito convention of stochastic calculus; however we avoid this discussion by working directly with the Fokker-Planck equation. Notice that when $T = 0$, Eq. (S16) is the transport equation for Eq. (S13).

Now, if the potential is periodic with period λ , we can use the well known Lifson-Jackson formula for the late time effective diffusion constant for Langevin processes in a potential and with varying diffusivity, both of period λ :

$$D^* = \frac{\lambda^2}{\int_0^\lambda dq \frac{\exp[\beta\phi(q)]}{D_e(q)} \int_0^\lambda dq \exp[-\beta\phi(q)]}. \quad (\text{S17})$$

In this case, we can easily show by integration by parts that

$$D^*(\kappa) = \frac{k_B T \lambda^2}{\gamma_{11} \int_0^\lambda dq \exp[-\beta\phi(q)] \int_0^\lambda dq \exp[\beta\phi(q)] \left(1 - \frac{\kappa\gamma_{12}^2\phi''(q)}{k_B T \lambda^2 \gamma_{11}^2} \right)}. \quad (\text{S18})$$

We thus see that in the regime where κ is small, the effect of a finite value of κ is to increase the long time diffusivity of the particle. We can also write $\phi(q) = \Delta U \psi(\frac{q}{\lambda})$, where ΔU is an adjustable energy scale (for instance, it could be related to the laser intensity and in which case λ would be a wavelength). This allows us to write

$$D^*(\kappa) = \frac{k_B T}{\gamma_{11} \int_0^1 dp \exp[-\beta\Delta U \psi(p)] \int_0^1 dp \exp[\beta\Delta U \psi(p)] \left(1 - \frac{\kappa\Delta U^2 \psi''(p)\gamma_{12}^2}{k_B T \lambda^2 \gamma_{11}^2} \right)}. \quad (\text{S19})$$

Therefore, even if κ is very small, increasing the value of ΔU or decreasing the value of L can be used to make the change in the long time diffusion constant measurable.

In the absence of any compliance, $\kappa = 0$, and no potential we note that

$$D^*(0) = \frac{k_B T}{\gamma_{11}}, \quad (\text{S20})$$

this corresponds to the modification of the diffusivity of q_1 when the *surface* q_1 is rigid but still exerts a hydrodynamic interaction.

From a physical standpoint, the fluctuations of q_2 can only influence the dynamics of q_1 on the time scale τ_2 before they are averaged out. However, if the typical time that q_1 takes to climb over the potential barrier to the next period is of order τ_2 , then q_2 can effect the long time diffusion of q_1 . In this way, fast and small degrees of freedom coupled to a slower degree of freedom can be detected even at low temporal resolution, provided one has sufficiently good statistics. By changing variables, we obtain

$$\frac{D^*(\kappa)}{D^*(0)} = \left(1 - \frac{\int_0^1 dp \exp[\beta\Delta U \psi(p)] \frac{\kappa\Delta U^2 \psi''(p)\gamma_{12}^2}{k_B T \lambda^2 \gamma_{11}^2}}{\int_0^1 dp \exp[\beta\Delta U \psi(p)]} \right)^{(-1)}. \quad (\text{S21})$$

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Now, setting

$$v = \beta \Delta U \quad (\text{S22})$$

gives

$$\frac{D^*(\kappa)}{D^*(0)} = \left(1 - \frac{\kappa k_B T v^2 \gamma_{12}^2}{\lambda^2 \gamma_{11}^2} \frac{\int_0^1 dp \exp[v\psi(p)] \psi'^2(p)}{\int_0^1 dp \exp[v\psi(p)]} \right)^{(-1)}. \quad (\text{S23})$$

If we take

$$\psi(u) = \sin(2\pi u), \quad (\text{S24})$$

we find

$$\frac{D^*(\kappa)}{D^*(0)} = \left(1 - \frac{4\pi^2 \kappa k_B T \gamma_{12}^2}{\lambda^2 \gamma_{11}^2} \frac{v I_1(v)}{I_0(v)} \right)^{(-1)}, \quad (\text{S25})$$

where $I_n(z)$ denotes the modified Bessel function of the first kind.

This can be written as

$$\frac{D^*(\kappa)}{D^*(0)} \simeq (1 + \alpha), \quad (\text{S26})$$

where

$$\alpha = \frac{4\pi^2 \kappa k_B T \gamma_{12}^2}{\lambda^2 \gamma_{11}^2} \frac{v I_1(v)}{I_0(v)}. \quad (\text{S27})$$

In the next section, we will see that it is natural to write

$$\alpha = 4\pi^2 \epsilon \frac{\gamma_{12}^2}{\gamma_{11}^2} \frac{v I_1(v)}{I_0(v)}, \quad (\text{S28})$$

where

$$\epsilon = \frac{\kappa k_B T}{\lambda^2}, \quad (\text{S29})$$

is the effective dimensionless compliance parameter that must be small for the above analysis to be valid.

The enhancement of the relative diffusion constant is clearly maximized by choosing small values of λ so that the variable q_2 can influence the jumping rate between neighboring minima rather than being averaged out if λ were large. We also see that the relative change depends on the function of the energy scale of the periodic potential

$$g(v) = \frac{v I_1(v)}{I_0(v)}, \quad (\text{S30})$$

which is a monotonically increasing function of v . This means that the relative change is increased with the energy scale ΔU of the trap.

The Fokker-Planck equation in Eq. (S16) corresponds to the Ito stochastic differential equation

$$\frac{dq_1}{dt} = -\mu_e(q_1)\phi'(q_1) + T\mu_e'(q_1) + \sqrt{2T\mu_e(q_1)}\eta(t), \quad (\text{S31})$$

where $\eta(t)$ is a zero-mean white noise with correlation function $\langle \eta(t)\eta(t') \rangle = \delta(t - t')$.

In the next section, we will see that this process has exactly the same diffusion constant as the process q_1 in the original hydrodynamically-coupled equations (S3 S4). Clearly, Eq. (S31) represents a temporal coarse graining of the original process q_1 on time scales greater than τ_2 . Finally, at very short time scales $t < \tau_2$, the microscopic time diffusion constant can similarly be deduced from Eq. (S3) and is given by

$$D_m = k_B T \mu_{11} = \frac{k_B T \gamma_{22}}{\gamma_{11} \gamma_{22} - \gamma_{12}^2}. \quad (\text{S32})$$

The expression D_m is simply the diffusion constant for the tracer in the absence of any hydrodynamic interactions with q_2 and no periodic potential; or, said otherwise, the bulk diffusivity of q_1 .

III. SUPPLEMENTARY NOTE 3: CORRECTION TO THE DIFFUSION BASED ON EXACT FORMULA FOR DIFFUSIVITY AND HOMOGENISATION THEORY

For small κ , q_2 is dominated by its fluctuations which scale *via* the equipartition of energy as

$$\delta q_2 \sim \sqrt{\kappa k_B T} . \quad (\text{S33})$$

Therefore, from now on, we will use the dimensionless position variables

$$q_1 = \lambda q \text{ and } q_2 = \sqrt{\kappa k_B T} u , \quad (\text{S34})$$

and again we also write $\beta\phi(q_1) = v\psi(\frac{q_1}{\lambda})$. The coupled Langevin equations can be rewritten in terms of the dimensionless time

$$s = \frac{\mu_{11} k_B T}{\lambda^2} t \quad (\text{S35})$$

to yield

$$\frac{dq}{ds} = -v\psi'(q) - \frac{\mu_{12}}{\mu_{11}\sqrt{\epsilon}} u + \eta'_1(s) \quad (\text{S36})$$

$$\frac{du}{ds} = -\frac{\mu_{22}}{\mu_{11}} \frac{u}{\epsilon} - v \frac{\mu_{12}}{\mu_{11}\sqrt{\epsilon}} \psi'(q) + \frac{1}{\sqrt{\epsilon}} \eta'_2(s) , \quad (\text{S37})$$

where $\eta'_1(s)$ and $\eta'_2(s)$ are Gaussian white noises with correlation functions

$$\langle \eta'_i(s) \eta'_j(s') \rangle = 2 \frac{\mu_{ij}}{\mu_{11}} \delta(s - s') . \quad (\text{S38})$$

and ϵ is defined in Eq. (S29).

The late time diffusion coefficient for q_1 is then given by via

$$2D^* t \simeq \langle q_1^2(t) \rangle = \lambda^2 \langle q^2(s) \rangle \simeq 2\lambda^2 D_q^* s = 2D_q \mu_{11} k_B T t . \quad (\text{S39})$$

This means that the late time diffusion constant for q_1 , D^* , is related to the late time diffusion constant of the scaled process q by

$$D^* = \mu_{11} k_B T D_q^* \left(\epsilon, v, \frac{\mu_{22}}{\mu_{11}}, \frac{\mu_{12}}{\mu_{11}} \right) , \quad (\text{S40})$$

where we have given the explicit functional dependencies for D_q^* . We now carry out a multiscale analysis along the lines of [12] where the correction of diffusion of a Brownian particle in a periodic potential due to inertial effects was given. Here, we will use the Stokes-Einstein relation to compute the diffusion constant and add an additional force h acting on the variable q in addition to the periodic potential. The Fokker-Planck equation for the variables q and u is then given by

$$\frac{\partial P}{\partial s} = H P , \quad (\text{S41})$$

where H is

$$H P = \left(H_0 + \frac{1}{\sqrt{\epsilon}} H_1 + \frac{1}{\epsilon} H_2 \right) P , \quad (\text{S42})$$

and the corresponding terms are given by

$$H_0 P = \frac{\partial}{\partial q} \left(\frac{\partial P}{\partial q} + [v\psi'(q) - h] P \right) \quad (\text{S43})$$

$$H_1 P = \frac{\mu_{12}}{\mu_{11}} \left[u \frac{\partial P}{\partial q} + 2 \frac{\partial^2 P}{\partial q \partial u} + (v\psi'_1(q) - h) \frac{\partial P}{\partial u} \right] \quad (\text{S44})$$

$$H_2 P = \frac{\mu_{22}}{\mu_{11}} \frac{\partial}{\partial u} \left(\frac{\partial P}{\partial u} + u P \right) . \quad (\text{S45})$$

The expansion we will make is based on small ϵ but also depends on the parameters μ_{22}/μ_{11} and μ_{12}/μ_{11} , the separation of scales needed to implement the expansion is more precisely states as

$$1 \ll \frac{1}{\sqrt{\epsilon}} \frac{\mu_{12}}{\mu_{11}} \ll \frac{1}{\epsilon} \frac{\mu_{22}}{\mu_{11}}, \quad (\text{S46})$$

and notably the second inequality implies that

$$\frac{\mu_{12}}{\mu_{11}} \ll \frac{1}{\sqrt{\epsilon}} \frac{\mu_{22}}{\mu_{11}}. \quad (\text{S47})$$

It will be useful later to employ the adjoint $H^\dagger$ of H given by

$$H^\dagger = H_0^\dagger + \frac{1}{\sqrt{\epsilon}} H_1^\dagger + \frac{1}{\epsilon} H_2^\dagger, \quad (\text{S48})$$

where

$$H_0^\dagger f = \frac{\partial^2 f}{\partial q^2} - (v\psi'(q) - h) \frac{\partial f}{\partial q}, \quad (\text{S49})$$

$$H_1^\dagger f = \frac{\mu_{12}}{\mu_{11}} \left(-u \frac{\partial f}{\partial q} + 2 \frac{\partial^2 f}{\partial q \partial u} - [v\psi'(q) - h] \frac{\partial f}{\partial u} \right), \quad (\text{S50})$$

$$H_2^\dagger f = \frac{\mu_{22}}{\mu_{11}} \left(\frac{\partial^2 f}{\partial u^2} - u \frac{\partial f}{\partial u} \right). \quad (\text{S51})$$

We proceed by computing the average velocity (denoted here by V) of q due to the applied field h for a tracer particle which is started with the local equilibrium distribution within a single period of the potential. This is given by

$$V = \left\langle \frac{dq}{ds} \right\rangle = \int_{-\infty}^{\infty} dq du du_0 \int_{\Lambda} dq_0 q \frac{dP(q, u|q_0, u_0; s)}{dt} P_{\text{peq}}(q_0, u_0), \quad (\text{S52})$$

where Λ the integration range over q_0 is over a single period of length 1 and $P_{\text{peq}}(q, u)$ is the equilibrium distribution for the variable u and the periodized variable $q^* = q \bmod(1)$. Using the Fokker-Planck equation (S41) then gives

$$V = \int_{-\infty}^{\infty} dq du du_0 \int_{\Lambda} dq_0 q [HP(q, u|q_0, u_0; s)] P_{\text{peq}}(q_0, u_0) = \int_{-\infty}^{\infty} dq du du_0 \int_{\Lambda} dq_0 [H^\dagger q] P(q, u|q_0, u_0; s) P_{\text{peq}}(q_0, u_0), \quad (\text{S53})$$

where

$$H^\dagger q = -v\psi'(q) + h - \frac{1}{\sqrt{\epsilon}} \frac{\mu_{12}}{\mu_{11}} u. \quad (\text{S54})$$

We notice that $H^\dagger q$ is periodic in q , so we can write

$$\int_{-\infty}^{\infty} du du_0 \int_{\Lambda} dq_0 P(q, u|q_0, u_0; s) P_{\text{peq}}(q_0, u_0) = \int_{-\infty}^{\infty} du du_0 \int_{\Lambda} dq dq_0 [H^\dagger q] \sum_{n=-\infty}^{\infty} P(q+n, u|q_0, u_0; s) P_{\text{peq}}(q_0, u_0). \quad (\text{S55})$$

However, $\sum_{n=-\infty}^{\infty} P(q+n, u|q_0, u_0; s) = P_p(q, u|q_0, u_0; s)$ is simply the transition density for the periodic variable q^* on Λ . Clearly $P_p(q, u|q_0, u_0; s)$ satisfies the same Fokker-Planck equation as P but restricted to $q \in \Lambda$ and with periodic boundary conditions.

As we have assumed that q^* starts in equilibrium we find have

$$\int_{-\infty}^{\infty} du_0 \int_{\Lambda} dq_0 P_p(q, u|q_0, u_0; s) P_{\text{peq}}(q_0, u_0) = P_{\text{peq}}(q_0, u_0), \quad (\text{S56})$$

and we thus obtain the Stratonovich formula

$$V = \int_{-\infty}^{\infty} du \int_{\Lambda} dq [H^\dagger q] P_{\text{peq}}(q, u), \quad (\text{S57})$$

where

$$HP_{\text{peq}}(q, u) = 0, \quad (\text{S58})$$

and has the periodic boundary condition $P_{\text{peq}}(q + 1, u) = P_{\text{peq}}(q, u)$. The Stokes-Einstein formula then gives the diffusion constant for the variable q as

$$D_q^* = \frac{\partial V}{\partial h} = \int_{-\infty}^{\infty} du \int_{\Lambda} dq \left[\frac{\partial H^\dagger}{\partial h} q \right] P_{\text{peq}0}(q, u) + [H^\dagger q] \frac{\partial}{\partial h} P_{\text{peq}}(q, u). \quad (\text{S59})$$

In Eq. (S59), everything is evaluated at $h = 0$, and the integral over q is on a single period Λ of the periodic potential ψ . We now denote

$$\frac{\partial}{\partial h} P_{\text{peq}}(q, u) = r(q, u), \quad (\text{S60})$$

which obeys

$$Hr(q, u) - \frac{\partial}{\partial q} P_{\text{peq}0}(q, u) - \frac{\mu_{12}}{\mu_{11}} \frac{1}{\sqrt{\epsilon}} \frac{\partial}{\partial u} P_{\text{peq}0}(q, u) = 0, \quad (\text{S61})$$

and must in addition obey the integral constraint, due to the conservation of probability,

$$\int dq du \frac{\partial}{\partial h} P_{\text{peq}}(q, u) = \int dq du r(q, u) = 0. \quad (\text{S62})$$

Now we note that Eq. (S54) gives

$$\frac{\partial H^\dagger}{\partial h} q = 1, \quad (\text{S63})$$

and consequently

$$D_q^* = \int_{-\infty}^{\infty} du \int_{\Lambda} dq P_{\text{peq}0}(q, u) + [H^\dagger q] r(q, u). \quad (\text{S64})$$

This simplifies to

$$D_q^* = 1 - \int dq du \left(v\psi'(q) + \frac{1}{\sqrt{\epsilon}} \frac{\mu_{12}}{\mu_{11}} u \right) r(q, u). \quad (\text{S65})$$

The above expression can also be derived using established results on the theory of transport processes [4], or using a general Kubo-type formalism [5, 6]. In principle, the partial differential equation Eq. (S61) can be numerically solved and the integral in Eq. (S65) can be evaluated using the solution to give a numerically exact result for D^* for all values of ϵ , and we will show how this can be done in section [V]. Here we compute the corrections due to finite κ perturbatively for small ϵ while assuming that the variables μ_{ij} are of the same order to validate the expansion we make. We work in the adjoint representation writing

$$r(q, u) = P_{\text{peq}0}(q, u)w(q, u), \quad (\text{S66})$$

which gives

$$H^\dagger w(q, u) + v\psi'(q) + \frac{u\mu_{12}}{\mu_{11}\sqrt{\epsilon}} = 0, \quad (\text{S67})$$

along with the integral constraint

$$\int_{-\infty}^{\infty} du \int_{\Lambda} dq P_{\text{peq}0}(q, u)w(q, u) = 0. \quad (\text{S68})$$

Setting $h = 0$ in Eq. (S54), we see that Eq. (S67) has a solution $w(q, u) = q + A$, where A is a constant. The integral constraint means that we must choose

$$A = -\langle q \rangle_{\text{eq}} = \int_{-\infty}^{\infty} du \int_{\Lambda} dq q P_{\text{peq}0}(q, u). \quad (\text{S69})$$

However, we must ensure periodicity in q . We thus write

$$w(q, u) = q - \langle q \rangle_{\text{eq}} + s(q, u) , \quad (\text{S70})$$

where $s(q, u)$ has the boundary condition

$$1 + s(1, u) = s(0, u) , \quad (\text{S71})$$

and obeys the equation

$$H^\dagger s(q, u) = 0 . \quad (\text{S72})$$

Plugging this into the formula for D_q^* , we find

$$D_q^* = 1 - \int_{-\infty}^{\infty} du \int_{\Lambda} dq P_{\text{peq0}}(q, u) \left(v\psi'(q) + \frac{1}{\sqrt{\epsilon}} \frac{\mu_{12}}{\mu_{11}} u \right) (q - \langle q \rangle_{\text{eq}} + s(q, u)) . \quad (\text{S73})$$

The shift due to the term $\langle q \rangle_{\text{eq}}$ gives zero contribution to the integral. The part of the integrand containing q can be integrated by parts (the second term giving zero) to give

$$D_q^* = \frac{1}{Z_-} \exp[-v\psi(1)] - \int dq du P_{\text{peq0}}(q, u) \left(v\psi'(q) + \frac{1}{\sqrt{\epsilon}} \frac{\mu_{12}}{\mu_{11}} u \right) s(q, u) , \quad (\text{S74})$$

and here we have explicitly taken $\Lambda = [0, 1]$. In what follows, we denote

$$Z_{\pm} = \int_0^1 dq \exp(\pm v\psi(q)) . \quad (\text{S75})$$

The first term in the integral on the right-hand side of Eq. (S74) can be integrated by parts to give the computationally useful formula

$$D_q^* = - \int_{-\infty}^{\infty} du \int_{\Lambda} dq P_{\text{peq0}}(q, u) \left(\frac{\partial s(q, u)}{\partial q} + \frac{1}{\sqrt{\epsilon}} \frac{\mu_{12}}{\mu_{11}} u s(q, u) \right) . \quad (\text{S76})$$

Now, we look for a solution of the form

$$s(q, u) = s_0(q, u) + \sqrt{\epsilon} s_1(q, u) + \epsilon s_2(q, u) + \sqrt{\epsilon}^3 s_3(q, u) + O(\epsilon^2) . \quad (\text{S77})$$

This yields, expanding to order ϵ ,

$$H_2^\dagger s_0 = 0 \quad (\text{S78})$$

$$H_1^\dagger s_0 + H_2^\dagger s_1 = 0 \quad (\text{S79})$$

$$H_0^\dagger s_0 + H_1^\dagger s_1 + H_2^\dagger s_2 = 0 \quad (\text{S80})$$

$$H_0^\dagger s_1 + H_1^\dagger s_2 + H_2^\dagger s_3 = 0 \quad (\text{S81})$$

$$H_0^\dagger s_2 + H_1^\dagger s_3 + H_2^\dagger s_4 = 0 . \quad (\text{S82})$$

The resulting diffusion constant is then given by

$$\begin{aligned} D_q^* &= - \frac{1}{\sqrt{\epsilon}} \int dq du P_{\text{peq0}}(q, u) \frac{\mu_{12}}{\mu_{11}} u s_0(q, u) - \int dq du P_{\text{peq0}}(q, u) \left[\frac{\partial s_0(q, u)}{\partial q} + \frac{\mu_{12}}{\mu_{11}} u s_1(q, u) \right] \\ &\quad - \sqrt{\epsilon} \int dq du P_{\text{peq0}}(q, u) \left[\frac{\partial s_1(q, u)}{\partial q} + \frac{\mu_{12}}{\mu_{11}} u s_2(q, u) \right] \\ &\quad - \epsilon \int dq du P_{\text{peq0}}(q, u) \left[\frac{\partial s_2(q, u)}{\partial q} + \frac{\mu_{12}}{\mu_{11}} u s_3(q, u) \right] , \end{aligned} \quad (\text{S83})$$

and for notational simplicity, we have not explicitly written the integration ranges which are $u \in (-\infty, \infty)$ and $q \in [0, 1]$. The equation (S79), $H_0^\dagger s_0(q, u) = 0$, shows that $s_0(q, u) = s_0(q)$ is independent of u . Note that in terms of

the parity with respect to u , the operators $H_0^\dagger$ and $H_2^\dagger$ are even and $H_1^\dagger$ is odd. From the structure of the equations, this means that $s_n(q, u)$ is even in u for n even and odd in u for n odd. As a consequence we find that

$$D_q^* = - \int dq du P_{\text{peq0}}(q, u) \left[\frac{\partial s_0(q, u)}{\partial q} + \frac{\mu_{12}}{\mu_{11}} u s_1(q, u) \right] - \epsilon \int dq du P_{\text{peq0}}(q, u) \left[\frac{\partial s_2(q, u)}{\partial q} + \frac{\mu_{12}}{\mu_{11}} u s_3(q, u) \right], \quad (\text{S84})$$

and it is easy to see that the higher order corrections are also integer powers of ϵ . We now consider Eq. (S79): $H_1^\dagger s_0 + H_2^\dagger s_1 = 0$. From this we see that solution for s_1 exists if the Fredholm alternative condition holds (the solvability condition)

$$\int du P_{\text{eq}}(u) H_1^\dagger s_0(q) = 0, \quad (\text{S85})$$

where $P_{\text{eq}}(u) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right)$, the marginal equilibrium probability distribution of u , is in the kernel of the adjoint of H_2 , that is to say $H_2^\dagger$. We find that

$$H_1^\dagger s_0(q) = -\frac{\mu_{12}}{\mu_{11}} u \frac{\partial s_0(q)}{\partial q}, \quad (\text{S86})$$

and this gives

$$s_1(q, u) = -\frac{\mu_{12}}{\mu_{22}} u \frac{\partial s_0(q)}{\partial q}. \quad (\text{S87})$$

Now, consider the equation Eq. (S80), $H_0^\dagger s_0 + H_1^\dagger s_1 + H_2^\dagger s_2 = 0$. Here, we find that

$$H_1^\dagger s_1 = \frac{\mu_{12}^2}{\mu_{11}\mu_{22}} \left([u^2 - 2] \frac{\partial^2 s_0(q)}{\partial q^2} + v\psi'(q) \frac{\partial s_0(q)}{\partial q} \right). \quad (\text{S88})$$

Integrating Eq. (S80) with respect to $P_{\text{eq}}(u)$ then gives the solvability condition for s_2 as

$$H_0^\dagger s_0 + \frac{\mu_{12}^2}{\mu_{11}\mu_{22}} \left(-\frac{\partial^2 s_0(q)}{\partial q^2} + v\psi'(q) \frac{\partial s_0(q)}{\partial q} \right) = 0, \quad (\text{S89})$$

which is simply

$$\left(1 - \frac{\mu_{12}^2}{\mu_{11}\mu_{22}} \right) H_0^\dagger s_0 = 0. \quad (\text{S90})$$

This has the solution

$$s_0(q) = A \int_0^q \exp[v\psi(p)] dp, \quad (\text{S91})$$

where A is a constant. Note that we can choose $s_0(0) = 0$ without loss of generality as D_q^* is independent of any constant term in $s(q)$. Using the boundary condition $s_0(1) = -1$ then gives

$$s_0(q) = -\frac{\int_0^q \exp[v\psi(p)] dp}{Z_+}, \quad (\text{S92})$$

and so

$$s'_0(q) = -\frac{\exp[v\psi(q)]}{Z_+}. \quad (\text{S93})$$

This then yields

$$s_1(q, u) = \frac{\mu_{12}}{\mu_{22}} u \frac{\exp[v\psi(q)]}{Z_+}. \quad (\text{S94})$$

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Importantly, we see that $s_1(q, u)$ is periodic in q and so the solution for s at order $\sqrt{\epsilon}$ obeys the full boundary condition *via* the term $s_0(q)$ solely, all the other corrections must consequently be periodic. This gives the $O(0)$ contribution to D_q^*

$$D_q^*(0) = \frac{1}{Z_+ Z_-} \left[1 - \frac{\mu_{12}^2}{\mu_{22} \mu_{11}} \right]. \quad (\text{S95})$$

Returning to Eq. (S80) $H_0^\dagger s_0 + H_1^\dagger s_1 + H_2^\dagger s_2 = 0$ we find that

$$\frac{\mu_{12}^2}{\mu_{11} \mu_{22}} [u^2 - 1] \frac{\partial^2 s_0(q)}{\partial q^2} + H_2^\dagger s_2 = 0. \quad (\text{S96})$$

This satisfies the solvability condition, and has solution

$$s_2(q, u) = u^2 \frac{\mu_{12}^2}{2\mu_{22}^2} \frac{\partial^2 s_0(q)}{\partial q^2} + s_{20}(q), \quad (\text{S97})$$

where $s_{20}(q)$ is undetermined (but must be periodic). Now solving Eq. (S81), $H_0^\dagger s_1 + H_1^\dagger s_2 + H_2^\dagger s_3$, we find the periodic solution

$$s_3(q, u) = -u \frac{\mu_{12}}{\mu_{22}} s'_{20}(q) + \frac{\mu_{12} u \exp[v\psi(q)] (\mu_{12}^2 u^2 v^2 \psi'(q)^2 + v\psi''(q) [6(\mu_{11} \mu_{22} - \mu_{12}^2) - u^2 \mu_{12}^2])}{6Z_+ \mu_{22}^3}. \quad (\text{S98})$$

To determine $s_{20}(q)$, we use the Fredholm alternative for solvability for Eq. (S82), $H_0^\dagger s_2 + H_1^\dagger s_3 + H_0^\dagger s_4 = 0$, which yields after some lengthy algebra

$$s_{20}(q) = \int_0^q \exp[v\psi(p)] dp \left[A' + \frac{\mu_{12}^2 (v^2 \psi'(p)^2 - v\psi''(p))}{2\mu_{22}^2 Z_+} \right], \quad (\text{S99})$$

where we have again taken $s_{20}(0) = 0$, and so the (periodic) boundary condition at $q = 1$ is $s_{20}(1) = 0$ (recalling that the inhomogeneous boundary condition is satisfied by $s_0(q)$), we thus have

$$A' Z_+ + \int_0^1 \exp[v\psi(p)] dp \frac{\mu_{12}^2 (v^2 \psi'(p)^2 - v\psi''(p))}{2\mu_{22}^2 Z_+} = 0, \quad (\text{S100})$$

and so

$$A' = -\frac{\mu_{12}^2}{\mu_{22}^2 Z_+^2} \int_0^1 \exp[v\psi(p)] v^2 \psi'(p)^2 dp. \quad (\text{S101})$$

This then yields

$$s'_{20}(q) = \left[-\frac{\mu_{12}^2}{\mu_{22}^2 Z_+^2} \int_0^1 \exp[v\psi(p)] v^2 \psi'(p)^2 dp + \frac{\mu_{12}^2 (v^2 \psi'(p)^2 - v\psi''(p))}{2\mu_{22}^2 Z_+} \right] \exp[v\psi(q)]. \quad (\text{S102})$$

From the above, we find that the correction to D_q^* to $O(\epsilon)$ is

$$\Delta D_{q^2}^* = \epsilon v^2 \frac{(\mu_{11} \mu_{22} - \mu_{12}^2)}{\mu_{22} \mu_{11}} \frac{\mu_{12}^2}{\mu_{22}^2} \frac{\int_0^1 \exp[v\psi(p)] \psi'(p)^2 dp}{Z_- Z_+^2}. \quad (\text{S103})$$

Using this to compute D^* then gives the formula derived from the heuristic argument presented in section (II).

IV. SUPPLEMENTARY NOTE 4: THE COMPLIANT LIMIT $\epsilon \gg 1$

In the limit $\epsilon \gg 1$, one can carry out a similar analysis to the limit $\epsilon \ll 1$. The Kubo-type formulae derived in the previous section still hold and the equation for $s(q, u)$ is now written as

$$[\epsilon H_0 + \sqrt{\epsilon} H_1 + H_2] s(q, u), \quad (\text{S104})$$

and the formula for D_q^* is unchanged.

Here, we find that one need to write a perturbative expansion of the form

$$s(q, u) = \sqrt{\epsilon} s_0(q, u) + s_1(q, u) + \frac{1}{\sqrt{\epsilon}} s_2(q, u) + O\left(\frac{1}{\epsilon}\right). \quad (\text{S105})$$

We find that

$$H_0^\dagger s_0 = 0 \quad (\text{S106})$$

$$H_0^\dagger s_1 + H_1^\dagger s_0 = 0 \quad (\text{S107})$$

$$H_0^\dagger s_2 + H_1^\dagger s_1 + H_2^\dagger s_0 = 0. \quad (\text{S108})$$

The equation $H_0 s_0 = 0$ then gives $s_0(q, u) = s_0(u)$, and so, to leading order, we find

$$D_q^* = - \int_{-\infty}^{\infty} du \int_{\Lambda} dq P_{\text{eq}0}(q, u) \left(\frac{\partial s_1(q, u)}{\partial q} + \frac{\mu_{12}}{\mu_{11}} s_0'(u) \right) + O\left(\frac{1}{\sqrt{\epsilon}}\right). \quad (\text{S109})$$

We impose the non-periodic boundary condition on the $O(1)$ term $s_1(q, u)$

$$s_1(0, u) = 0; \quad 1 + s_1(1, u) = 0. \quad (\text{S110})$$

The term s_1 obeys the equation $H_1 s_0 + H_0 s_1 = 0$, where $H_1 s_0 = -\frac{\mu_{12}}{\mu_{11}} v \psi'(q) s'(u)$. We notice that

$$\int_{\Lambda} dq P_{\text{eq}}(q) H_1 s_0 = 0, \quad (\text{S111})$$

where

$$P_{\text{eq}}(q) = \frac{\exp[-v \psi(q)]}{Z_-}, \quad (\text{S112})$$

and so this equation obeys the solvability criterion. The solution is given by

$$s_1(q, u) = -q \frac{\mu_{12} s_0'(u)}{\mu_{11}} + \frac{1}{Z_+} \left[\frac{\mu_{12} s_0'(u)}{\mu_{11}} - 1 \right] \int_0^q dp \exp[v \psi(p)]. \quad (\text{S113})$$

Using this result in Eq. (S109) then gives to leading order

$$D_q^* = \frac{1}{Z_+ Z_-} \left[1 - \frac{1}{\sqrt{2\pi}} \frac{\mu_{12}}{\mu_{11}} \int_{-\infty}^{\infty} du \exp\left(-\frac{u^2}{2}\right) s_0'(u) \right] + O\left(\frac{1}{\sqrt{\epsilon}}\right). \quad (\text{S114})$$

Finally, the equation for $s_0(u)$ can be found by using the solvability condition for s_2

$$0 = \int_{\Lambda} dq P_{\text{eq}}(q) \left[H_0^\dagger s_2 + H_1^\dagger s_1 + H_2^\dagger s_0 \right] = H_2^\dagger s_0(u) + \int_{\Lambda} dq P_{\text{eq}}(q) H_1^\dagger s_1(q, u). \quad (\text{S115})$$

After some algebra, the solution which is integrable with respect to the equilibrium measure $P_{\text{eq}}(u)$ for u is

$$s_0(u) = \frac{u \mu_{12} \mu_{11}}{(\mu_{11} \mu_{22} - \mu_{12}^2) Z_+ Z_- + \mu_{12}^2}. \quad (\text{S116})$$

Using this in Eq. (S114) then gives

$$D_q^* = \frac{\mu_{11} \mu_{22} - \mu_{12}^2}{Z_+ Z_- (\mu_{11} \mu_{22} - \mu_{12}^2) + \mu_{12}^2} + O\left(\frac{1}{\sqrt{\epsilon}}\right). \quad (\text{S117})$$

Note that when there is no periodic potential ($v = 0$) one has $Z_+ = Z_- = 1$, and we find

$$D_q^*(\epsilon = \infty) = k_B T \frac{\mu_{11} \mu_{22} - \mu_{12}^2}{\mu_{22}} = \frac{k_B T}{\gamma_{11}} \quad (\text{S118})$$

which is the same result for the diffusion constant in the non-compliant case $\epsilon = 0$ and with $v = 0$. In terms of the unscaled variables this then gives

$$D^* = k_B T \mu_{11} \frac{\mu_{11} \mu_{22} - \mu_{12}^2}{\lambda^{-2} Z_+ Z_- (\mu_{11} \mu_{22} - \mu_{12}^2) + \mu_{12}^2} + O\left(\frac{1}{\sqrt{\epsilon}}\right). \quad (\text{S119})$$

V. SUPPLEMENTARY NOTE 5: NUMERICAL COMPUTATION OF D^* FROM THE KUBO FORMULA

The diffusion constant D_q^* related to D^* by Eq. (S40) can be numerically computed by solving Eq. (S61) with periodic boundary conditions and the integral constraint Eq. (S62). This can be carried out by using the FlexPDE Multiphysics Software For Partial Differential Equations. The numerical domain is finite and taken over $[-1/2, 1/2]$ for the variable q with periodic boundary conditions. The variable u is unbounded, but we take a finite domain with $u \in [-L, L]$ (typically, it suffices to take $L = 5$), and impose no-flux boundary conditions at the surfaces $u = \pm L$. This means that the effective potential for u becomes infinite at the edges, but in practice the variable $r(q, u)$ has decayed to zero at these values of u . Examples are shown in Figs. 4, S1, and S2, for various parameters.

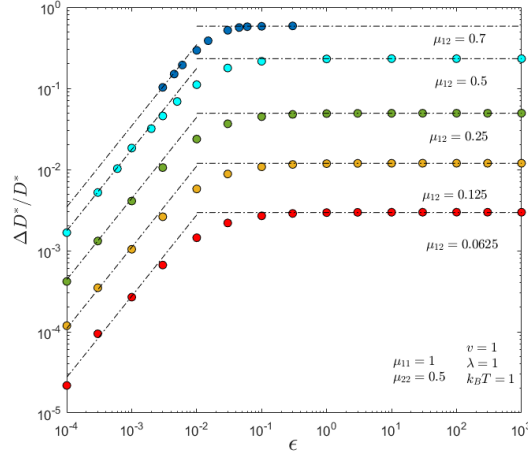


FIG. S1. Relative increase $\Delta D^*/D^* = [D^*(\epsilon) - D^*(0)]/D^*(0)$ of the late-time diffusion coefficient D^* of the test particle as a function of the dimensionless compliance ϵ , for $\mu_{11} = 1.0$, $\mu_{22} = 0.5$, $v = 1.0$, $\lambda = 1.0$, $k_B T = 1.0$, and various μ_{12} as indicated. The symbols correspond to the Kubo formula of Eq. (S65). The dash-dotted lines correspond to the small- ϵ prediction of Eq. (5), and the large- ϵ prediction of Eq. (11).

VI. SUPPLEMENTARY NOTE 6: HEURISTIC ARGUMENT FOR MODIFICATION OF DIFFUSION WITH INERTIA

Here, we show how the heuristic method proposed beforehand can be used to compute the effective diffusion constant for a particle due to first order inertial effects, recovering the result of [12]. This calculation also demonstrates how a periodic potential can be applied to detect the signature of inertia on late-time diffusion.

We consider the zero-temperature equation

$$m \frac{dv}{dt} + \gamma v = -\phi'(q), \quad (\text{S120})$$

where $v = \frac{dq}{dt}$ is the particle's velocity.

In operator notation, this formally gives

$$v(t) = -\frac{1}{\gamma} \left(1 + \tau \frac{d}{dt} \right)^{-1} \phi'(q), \quad (\text{S121})$$

where $\tau = \frac{m}{\gamma}$ is the short timescale over which the velocity relaxes and which would be difficult to resolve experimentally.

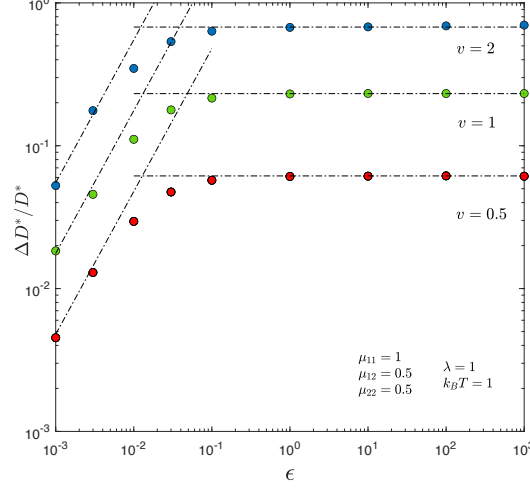


FIG. S2. Relative increase $\Delta D^*/D^* = [D^*(\epsilon) - D^*(0)]/D^*(0)$ of the late-time diffusion coefficient D^* of the test particle as a function of the dimensionless compliance ϵ , for $\mu_{11} = 1.0$, $\mu_{12} = 0.5$, $\mu_{22} = 0.5$, $\lambda = 1.0$, $k_B T = 1.0$, and various v as indicated. The symbols correspond to the Kubo formula of Eq. (S65). The dash-dotted lines correspond to the small- ϵ prediction of Eq. (5), and the large- ϵ prediction of Eq. (11).

Now, expanding to first order in τ , we find

$$v(t) = -\frac{1}{\gamma}\phi'(q) + \frac{\tau}{\gamma}\phi''(q)v(t), \quad (\text{S122})$$

and so

$$\frac{dq}{dt} = v(t) = -\mu_e(q)\phi'(q), \quad (\text{S123})$$

where

$$\mu_e(q) = \frac{1}{\gamma \left[1 - \frac{\tau}{\gamma}\phi''(q) \right]}. \quad (\text{S124})$$

The diffusion constant is thus

$$D_e(q) = \frac{k_B T}{\gamma \left[1 - \frac{\tau}{\gamma}\phi''(q) \right]}. \quad (\text{S125})$$

Now, applying the Lifson-Jackson formula, we obtain

$$D^* = \frac{k_B T \lambda^2}{\gamma \int_0^\lambda dq \exp[\beta\phi(q)] \left[1 - \frac{m}{\gamma^2}\phi''(q) \right] \int_0^\lambda dq \exp[-\beta\phi(q)]}. \quad (\text{S126})$$

This can be rewritten as

$$D^* = \frac{k_B T \lambda^2}{\gamma \int_0^\lambda dq \exp[\beta\phi(q)] \left[1 + \frac{m}{k_B T \gamma^2}\phi''(q) \right] \int_0^\lambda dq \exp[-\beta\phi(q)]}, \quad (\text{S127})$$

or to the same order

$$D^* = \frac{k_B T \lambda^2}{\gamma \int_0^\lambda dq \exp[\beta\phi(q)] \int_0^\lambda dq \exp[-\beta\phi(q)]} \left(1 - \frac{m}{k_B T \gamma^2} \frac{\int_0^\lambda dq \exp[\beta\phi(q)] \phi'(q)^2}{\int_0^\lambda dq \exp[\beta\phi(q)]} \right), \quad (\text{S128})$$

which recovers the result given in [1, 2] via multiscale analysis. Notice that in this particular case, the effect of inertia is to reduce the late time diffusion constant, and again, it is necessary to impose an external periodic potential to produce a measurable effect on the late time diffusivity.

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Enhanced dispersion of active microswimmers in confined flows

Marc Lagoin^{a,1}, Juliette Lacherez^{a,2}, Guirec de Tournemire^{a,2} , Ahmad Badr^{a,2}, Yacine Amarouchene^{a,1} , Antoine Allard^{a,1} , and Thomas Salez^{a,1}

Affiliations are included on p. 5.

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In the presence of a laminar shear flow, the diffusion of passive colloidal particles is enhanced in the direction parallel to the flow. This classical phenomenon is known as Taylor–Aris dispersion. Besides, microorganisms, such as active microswimmers, exhibit an effective diffusive behavior at long times. Combining the two ingredients above, a natural question then emerges on how the effective diffusion of active microswimmers is altered in shear flows—a widespread situation in natural environments with practical implications, e.g., regarding biofilm formation. In this Letter, we investigate the motility and dispersion of *Chlamydomonas reinhardtii* microalgae, within a rectangular microfluidic channel subjected to a sinusoidal Poiseuille flow. Using high-resolution optical microscopy and a particle-tracking algorithm, we reconstruct individual trajectories in various flow conditions and statistically analyze them through moment theory and sliding windowed demodulation. We find that the velocity fluctuations and the dispersion coefficient increase as the flow amplitude is increased, with only weak dependencies on the flow periodicity. Importantly, our results demonstrate that the generalization of Taylor–Aris law to active particles is valid.

microswimmers | algae | microfluidics | dispersion | shear flows

Microscopic organisms frequently exceed the length scale at which thermal Brownian motion alone can mediate efficient environmental exploration. To circumvent this constraint, they employ diverse active motility strategies in viscous media (1), including crawling (2), gliding (3, 4), and swimming (5–7). Among microswimmers, propulsion mechanisms exhibit notable variations: for instance, *Escherichia coli* functions as a pusher, using rotary motor-driven flagella for propulsion (8); whereas the biflagellate alga *Chlamydomonas reinhardtii*, from the *Volvox* family (9), acts as a puller relying on synchronous beating of its flagella (5). Diverse external stimuli, such as light for photosynthesis (10–13), or chemical gradients (14), are essential in determining the dynamics of microswimmers. Similarly, hydrodynamic interactions with the environment are fundamental. As a result of these, swimming microorganisms manifest a rich repertoire of behaviors, including effective viscosity (15), accumulation near boundaries (16–19), entrainment (20), upstream swimming (21, 22), Bretherton–Jeffery orbits (23), resonant transverse swimming (24, 25), and emergent interactions (26, 27). Despite the persistent self-propelled motion of microswimmers, their long-time dynamics often resembles the one of passive Brownian particles, which is characterized through an effective diffusion process (28–32).

In another context, the seminal studies of Taylor (33–35) and Aris (36) established that shear flows can significantly enhance the dispersion of passive microscopic solutes—a now-classical effect known as Taylor–Aris dispersion. Further works have extended this framework to incorporate the effects of channel geometry (37), porosity (38–40), reactivity (41), flow complexity (42–45), short-time distribution (46), and open or charged boundaries (47). The underlying coupling between advection and diffusion has remained a vibrant topic of theoretical and numerical inquiry (48, 49). The Taylor–Aris mechanism has also found impactful applications, e.g., in particle-size characterization (50–54).

Given that active microswimmers effectively diffuse at long times, and often navigate within external flows in natural environments, a compelling question arises: does an analogue of the Taylor–Aris law govern their transport? While theoretical and computational models have proliferated on active-particle dispersion in external flows (21, 45, 55–58), and on the influence of oscillatory flows on active suspensions (59), experimental confirmation remains scarce, primarily due to challenges in recording the dynamics over extended time scales. Nevertheless, recent and preliminary experimental studies addressed *E. coli* in flows (60), bacterial scattering in microfluidic lattices (61), as

Significance

This study introduces a unified approach combining microfluidics, Langevin simulations, and statistical analysis to examine how *Chlamydomonas reinhardtii*, an active swimming microalgae, disperses in confined flows. The findings extend classical Taylor–Aris dispersion theory—originally for passive Brownian particles—to active swimmers, revealing a transition from ballistic to diffusive behaviors influenced by the flow. Shear flows are shown to impact the swimmers' dynamics nonlinearly, enhancing spreading in complex ways. These results provide critical insight into microswimmers' behavior in fluid environments, relevant to both artificial and natural settings. The work has broader implications for microbial ecology, suggesting ways to understand processes like biofilm formation, nutrient transport, and pathogen dispersal in ecosystems where fluid dynamics shape microorganism movement and interactions.

The authors declare no competing interest.

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¹To whom correspondence may be addressed. Email: marc.lagoin@u-bordeaux.fr, yacine.amarouchene@u-bordeaux.fr, antoine.allard@u-bordeaux.fr, or thomas.salez@cnrs.fr.²J.L., G.d.T., and A.B. contributed equally to this work.This article contains supporting information online at <https://www.pnas.org/lookup/suppl/doi:10.1073/pnas.2519691122/-DCSupplemental>.

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well as dispersion of microalgae as gyrotactic swimmers (58, 59). In the present article, we quantitatively investigate the Taylor–Aris dispersion of model active particles, using a combination of experiments, numerical simulations, and theory.

Experiments

We employ the green alga *C. reinhardtii* CC409 as a model active swimmer, owing to its well-characterized motility and widespread use in microswimmer research (62). See *Materials and Methods* for details on the experiments. Further details on cell culture and experimental preparation are also provided in *SI Appendix*. This strain swims at a characteristic speed of approximately $V_0 \approx 100 \mu\text{m/s}$ (63), with a decorrelation time of around ~ 1 s, beyond which it exhibits an effective diffusion with a diffusion coefficient on the order of $D_0 \approx 10^{-9} \text{m}^2/\text{s}$, as obtained from 1,296 single-cell trajectories in the bulk and in the absence of external flow (*SI Appendix*).

To probe the long-time dynamics of active dispersion, we impose a time-periodic sinusoidal flow that maintains the algae within the field of view over a reasonable duration. Our experimental approach builds on theoretical frameworks describing active-particle dispersion in external flows (21) and the influence of oscillatory flows on passive thermal dispersion (42). Fig. 1 shows the experimental setup together with representative trajectories of individual cells under oscillatory microfluidic flow.

The fluid velocity field $\vec{u}(y, z, t)$ along the flow axis x (with unit vector $\vec{e}_x$), at time t , can thus be accurately described as a laminar, pressure-driven Poiseuille flow, expressed as

$$\vec{u}(y, z, t) = \frac{P}{R_h} f(y, z) \sin\left(\frac{2\pi}{T}t + \phi_0\right) \vec{e}_x, \quad [1]$$

where P is the oscillation amplitude of the pressure, T is the oscillation period, R_h is the reduced hydraulic resistance of the channel, $f(y, z)$ is the dimensionless cross-sectional profile and ϕ_0 is an arbitrary phase shift. For instance, in a two-dimensional case, one would have $f(y) = 4(y/W)(1 - y/W)$, with W the channel width, while for a three-dimensional parallelepipedic channel we refer to ref. 65.

The central dimensionless control parameter in our problem is the Péclet number Pe , that compares flow-induced advection to effective diffusion. It is defined as

$$Pe \equiv \frac{P}{\sqrt{2}R_h} \frac{W}{D_0}, \quad [2]$$

where $P/(\sqrt{2}R_h)$ corresponds to the RMS velocity of the oscillatory flow.

In addition, we assume that the position of a single cell can be expressed as

$$x(t) = \hat{x}(t) + A(y(t), t) \cos\left(\frac{2\pi}{T}t\right), \quad [3]$$

where $A(y(t), t)$ is an unknown, potentially random function, and $\hat{x}(t)$ represents the cell's position in the comoving frame associated with the flow. For a nonswimming cell, the velocity is expected to follow a sinusoidal pattern, mirroring the flow profile. If the particle is placed at a fixed transverse position y , the amplitude of the oscillation is determined by the local flow velocity at that position. However, in the case of an active particle that randomly explores both x and y , the amplitude of the position oscillation inherited from the flow becomes modulated by the particle's stochastic exploration along the y -axis.

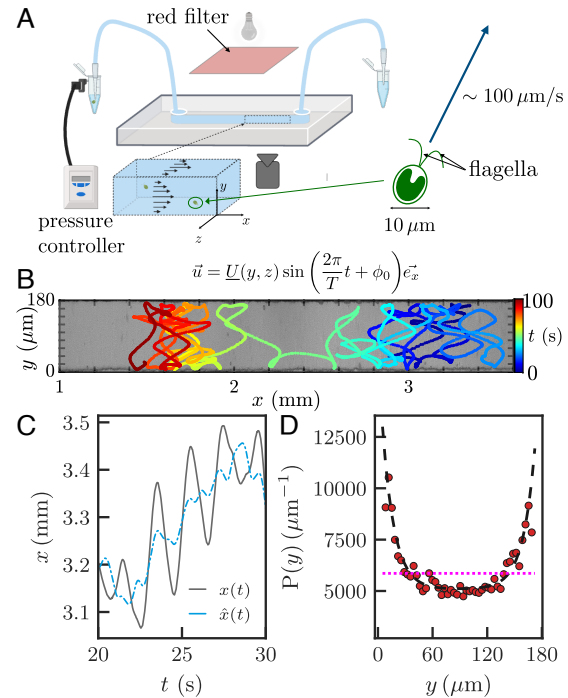


Fig. 1. Dynamics of microalgae in an oscillatory microfluidic flow. (A) Schematic of the experimental setup. A suspension of microalgae is confined in a microfluidic channel with rectangular cross-section. A pressure controller at one end generates a sinusoidal Poiseuille flow, while a constant pressure is applied at the other end via a fluid reservoir. The algae are illuminated with red-filtered light and imaged using a high-speed camera (equipped with a $\times 5$ objective). (B) Representative top-view image of the channel ($\times 5$ magnification; frame rate: 40 Hz; exposure time: 1 ms) with a typical trajectory superimposed. The colormap encodes time along the trajectory. (C) Slice of a time series of the longitudinal position $x(t)$. The demodulated signal $\hat{x}(t)$ is overlaid. (D) Probability density function $P(y)$ of the transverse position y , as estimated from 27 trajectories exceeding 20 s in duration. The magenta dotted line indicates a uniform distribution. The black dashed line shows the best fit to Eq. 4. In panels (B–D), the experimental parameters are Péclet number $Pe = 9.15$ (i.e., $P = 1$ mbar), and flow-oscillation period $T = 2$ s.

Results

Trajectories and Probability Density Function. From Eq. 3, we use sliding windowed demodulation to experimentally extract the process $\hat{x}(t)$. Examples of $x(t)$ and $\hat{x}(t)$ are shown in Fig. 1C. In Fig. 1B, we observe that an individual cell explores the full channel width along y multiple times within a single trajectory. This is a condition required to apply Taylor–Aris results. Furthermore, the probability density function $P(y)$ of the transverse position y , aggregated over all trajectories under identical conditions, exhibits a pronounced nonuniformity, as shown in Fig. 1D. This form is well captured by the cycloidal distribution predicted for active particles (21):

$$P(y) = \mathcal{A} + \frac{\mathcal{B}}{2} \cosh\left[\lambda\left(y - \frac{W}{2}\right)\right], \quad \int dy P(y) = 1, \quad [4]$$

where $\mathcal{A}$, $\mathcal{B}$ and λ are three constants that depend on the geometry and the effective diffusion of the active process. Fit results yield: $\mathcal{A} \approx 5.1 \cdot 10^3 \text{m}^{-1}$ which represents the bulk density, and $\mathcal{B} \approx 25 \text{m}^{-1}$ which represents the strength of accumulation near the walls. In particular, $\lambda^{-1} \approx 13 \mu\text{m}$, which is about the size of

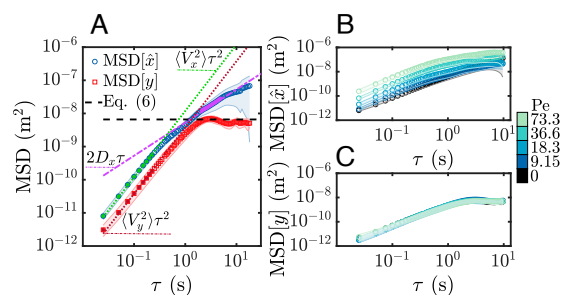


Fig. 2. MSD of microalgae under oscillatory microfluidic flow. (A) Experimental ensemble-averaged MSDs $\text{MSD}[\hat{x}](\tau)$ (circles) and $\text{MSD}[\hat{y}](\tau)$ (squares), respectively parallel and perpendicular to the flow, as functions of time increment τ , for Péclet number $\text{Pe} = 9.15$, and flow-oscillation period $T = 2$ s. A total of 27 trajectories, longer than 20 s each, are included in the average. Green, red, and purple symbol fillings indicate the time intervals used for estimating $\langle V_x^2 \rangle$, $\langle V_y^2 \rangle$, and D_x , respectively, and the associated dotted and dash-dotted lines indicate the ballistic and diffusive-like asymptotic behaviors, as indicated. The black dashed line corresponds to Eq. 6, using the best-fit parameters obtained from the fit of the transverse distribution $P(y)$ shown in Fig. 1D. (B) Ensemble-averaged $\text{MSD}[\hat{x}](\tau)$ for various Péclet numbers (see legend in next panel). (C) Ensemble-averaged $\text{MSD}[\hat{y}](\tau)$ for various Péclet numbers, as indicated.

one cell (7% of the channel width) describes the typical size of the boundary layer close to the walls. We further assume that hydrodynamic interactions with the channel walls introduce only higher-order corrections, with axial transport primarily dictated by the bulk advective flow, in line with the Taylor–Aris dispersion framework. The validity of this assumption is demonstrated in *SI Appendix*.

Mean-Square Displacements. To analyze the stochastic dynamics of the cell trajectories, we consider the mean square displacements (MSDs) of the processes $\hat{x}$ and $\hat{y}$ at time scale τ . The MSDs are the variances of the position increments at time scale τ . For instance, for $\hat{x}$, the MSD is defined as

$$\text{MSD}[\hat{x}](\tau) \equiv \overline{\delta_\tau \hat{x}^2} - \overline{\delta_\tau \hat{x}}^2, \quad \delta_\tau \hat{x}(t) \equiv \hat{x}(t + \tau) - \hat{x}(t), \quad [5]$$

where $\overline{}$ denotes the temporal average. In Fig. 2A, we present typical experimental measurements of $\text{MSD}[\hat{x}]$ and $\text{MSD}[\hat{y}]$ for fixed flow conditions. Asymptotically, for time scales much smaller than the decorrelation time $\tau_x \approx 1$ s in the x direction, the parallel MSD exhibits a ballistic scaling, i.e., $\text{MSD}[\hat{x}](\tau \ll \tau_x) \simeq \langle V_x^2 \rangle \tau^2$, where $\langle V_x^2 \rangle$ is the mean square velocity. Conversely, for time scales much larger than τ_x , the dynamics is diffusive-like, i.e., $\text{MSD}[\hat{x}](\tau \gg \tau_x) \simeq 2D_x \tau$, where D_x is the so-called dispersion coefficient. The latter coefficient encompasses in a general fashion the effects of i) the raw effective diffusion due to activity and ii) the possible Taylor–Aris mechanism at stake in this study. Besides, $\text{MSD}[\hat{y}]$ also exhibits a ballistic behavior at short times, meaning that the algae swim both parallel and perpendicular to the flow. However, we cannot observe a long-time diffusive-like behavior perpendicular to the flow, as $\text{MSD}[\hat{y}]$ saturates to a plateau at large times due to the presence of the walls, which imposes $y \in [0, W]$. Indeed, we find good agreement when comparing the plateau value of $\text{MSD}[\hat{y}]$ with the prediction:

$$\text{MSD}[\hat{y}](\tau \rightarrow +\infty) = 2 \int dy y^2 P(y). \quad [6]$$

Moreover, in Fig. 2B, we observe that $\text{MSD}[\hat{x}]$ increases as Pe increases, which indicates that both the variance of the speed and

the dispersion coefficient grow with the Péclet number. In sharp contrast, this behavior is absent for $\text{MSD}[\hat{y}]$, as all the curves of Fig. 2C collapse together for any value of Pe . We conclude from this latter observation that in our experimental range of Pe , the flow does not disturb significantly the algae swimming mechanism itself.

Velocity Variance. We now focus on the evolution of the variance of the velocity (see details in *SI Appendix*) with the flow amplitude. In Fig. 3A, we plot the experimental $\langle V_x^2 \rangle$ and $\langle V_y^2 \rangle$ as functions of Pe^2 , for different flow-oscillation periods T . Our results indicate that the flow-oscillation period has no significant effect within our parameter range. This is expected since, by design of our study, the period of the oscillation was chosen to remain greater than the typical decorrelation time of the particle random motion (42). Also, we recover that $\langle V_y^2 \rangle$ remains unchanged by the flow amplitude. Interestingly, we observe that $\langle V_x^2 \rangle$ increases linearly with Pe^2 , which is consistent with speed composition. Indeed, for a Brownian particle in a two-dimensional Poiseuille flow with maximum speed $\underline{U}$, one can prove that $\langle V_x^2 \rangle / \langle V_y^2 \rangle = 1 + \beta \underline{U}^2 / \langle V_y^2 \rangle$, where $\beta = 8/90$ is a geometric factor (*SI Appendix*). Assuming that this analysis is valid for active particles at long times, in our three-dimensional configuration, and using the RMS flow speed relevant to our oscillatory case instead of $\underline{U}$, one obtains:

$$\frac{\langle V_x^2 \rangle}{\langle V_y^2 \rangle} = 1 + \beta \frac{\text{Pe}^2}{\text{Pe}_A^2}, \quad [7]$$

where we have introduced the usual, zero-flow, active Péclet number (28–32), $\text{Pe}_A \equiv (W/D_0)\sqrt{\langle V_y^2 \rangle} \approx 3.2$. In Fig. 3B, we

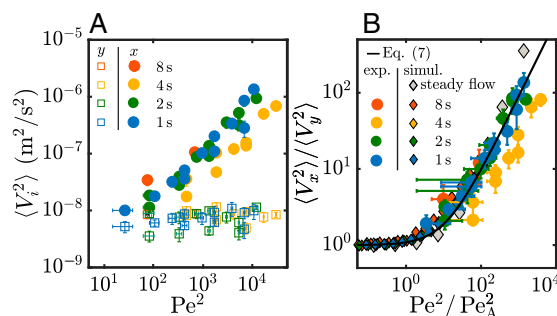


Fig. 3. Velocity fluctuations of microalgae under oscillatory microfluidic flow. (A) Experimental ensemble-averaged mean square velocities $\langle V_x^2 \rangle$ (circles) and $\langle V_y^2 \rangle$ (squares), respectively parallel and perpendicular to the flow, as functions of the squared Péclet number Pe^2 , for various flow-oscillation periods T , as indicated. (B) Ratio of the mean square velocities $\langle V_x^2 \rangle / \langle V_y^2 \rangle$ as a function of the squared ratio between Péclet number Pe and active Péclet number (28–32) $\text{Pe}_A \equiv (W/D_0)\sqrt{\langle V_y^2 \rangle} \approx 3.2$, for various flow-oscillation periods T , as indicated. The circles correspond to experimental data while the diamonds correspond to the results from two-dimensional Langevin-like simulations (21) in oscillatory flows, including one steady-flow reference case. The solid black line corresponds to Eq. 7 with $\beta = 8/90$ (*SI Appendix*). Each point represents an ensemble average over trajectories longer than 20 s. Vertical error bars denote one SD, while horizontal error bars account for uncertainties in the applied pressure (as specified by the manufacturer) and in the measurement of the baseline diffusion coefficient D_0 (*SI Appendix*).

confront the prediction of Eq. 7 to both the experimental and numerical data. We observe good agreement between theory, simulations, and experiments.

Dispersion coefficient. Let us finally investigate long-term dispersion. In Fig. 4A, we show the variation of the experimental dispersion coefficient D_x parallel to the flow direction as a function of the flow amplitude. We see no major systematic effect of the flow-oscillation period beyond error bars. Interestingly, we observe that D_x increases linearly with Pe^2 , which is consistent with the classical Taylor–Aris law for passive Brownian particles in steady viscous shear flows (33–36):

$$D_x = D_0 (1 + \alpha \text{Pe}^2), \quad [8]$$

where α is a geometric factor, that is equal to $1/210$ in two-dimensional Poiseuille flows, and where D_0 denotes the bulk diffusion coefficient of the considered particles. To test this mechanism further, we plot in Fig. 4B the ratio of the parallel dispersion coefficient D_x to the raw effective diffusion coefficient D_0 as a function of the squared Péclet number Pe^2 , for all the experimental and numerical data. The data are consistent with Eq. 8, hence revealing the main finding that the classical Taylor–Aris mechanism for dispersion of Brownian particles in shear flows can be extended to the case of model active swimmers exhibiting long-term effective diffusion. Two minor remarks should however be made. As a first remark, we measure $\alpha \equiv (D_x/D_0 - 1)/\text{Pe}^2 \approx (1.6 \pm 2.3) \cdot 10^{-3}$ from the experimental data, which, while being on the same order of magnitude, is systematically below the theoretical value $\alpha = 1/210 \approx 4.8 \cdot 10^{-3}$. This discrepancy may have several origins, beyond the obvious dimensionality difference which provides only minor corrections. First, by construction, long-time experimental data are more prone to statistical errors. Second, the cells may adapt to the flow in hardly predictable and testable ways. Third, D_x and D_0 are expected to conjointly vary from one living organism to the other, which implies that both should in principle be simultaneously measured for each given

individual cell. Unfortunately, it was not experimentally feasible to ensure that no cell entered or left the channel during one single experiment. It was therefore not possible to obtain the D_0 value corresponding to each D_x measurement. Instead, as already mentioned, we use the average effective diffusion coefficient measured on many algae, in the bulk, and in the absence of any flow (SI Appendix). As a second remark, we notice that the simulation results show a dependency with the flow period, that the experiments cannot confirm due to the current statistical resolution. Reducing this uncertainty would likely require an alternative experimental approach, such as tracking individual cells over multiple flow cycles and applying appropriate normalization to each trajectory.

Conclusion

In this article, we combined microfluidic experiments on *C. reinhardtii* microalgae, numerical Langevin simulations, and theory, in order to address the dispersion of model active microswimmers in viscous shear flows. Combining single-cell tracking and statistical analysis, we checked the expectation that algae preferentially accumulate near channel boundaries. Furthermore, in the direction parallel to the flow, we observed and quantified a ballistic-to-diffusive-like crossover, with a clear influence of the flow amplitude in both asymptotic regimes. We quantitatively rationalized all the parametric dependencies. Our central finding is that the classical Taylor–Aris law for dispersion of Brownian particles in shear flows can be extended to the case of model active microswimmers. This mechanism could have important practical implications in microorganism spreading and biofilm formation, within natural environments where flows are omnipresent. In future, it would be interesting to test the mechanism for model pushers rather than pullers, to incorporate external stimuli such as light or chemical cues, in order to bias the active Taylor–Aris dispersion uncovered here, and to test additional lubrication effects in stronger confinement. Moreover, we here used quasi-steady oscillatory flows to ensure sufficient statistical resolution, but increasing further the oscillation frequency beyond the natural decorrelation rate (42) and investigating the resulting biomechanical change in swimming would be an interesting avenue.

Materials and Methods

The supporting materials for this manuscript are provided in SI Appendix. Videos and numerical routines can be found on the online repository: https://github.com/EMetBrown-Lab/Chlamy_taylor_aris/tree/main.

Microfluidic Channel. A microfluidic channel is fabricated via soft lithography in a polydimethylsiloxane (PDMS) chip. The channel has a rectangular cross-section with vertical height $H = 100 \mu\text{m}$ along z , and width $W = 180 \mu\text{m}$ along y , and the total length is $L = 88 \text{ mm}$ along x . A 0.5% w/v solution of Pluronic F-127 is used to prevent cell adhesion, following established protocols (19).

Algae Suspension. The chip is filled with a dilute suspension of algae at a concentration between 2 and 5×10^6 cells per milliliter, ensuring negligible cell–cell interactions. Concentrations are measured using a hemocytometer, as described in ref. 64 (see SI Appendix for further details).

Flow Control. A pressure controller (Fluigent LU-FEZ-0345) is used to apply a target pressure P at one end of the channel. It can be modulated between 0 and 345 mbar. The controller supports sinusoidal pressure modulations with accurate waveform for periods $T \gtrsim 0.5 \text{ s}$, and guarantees precision within

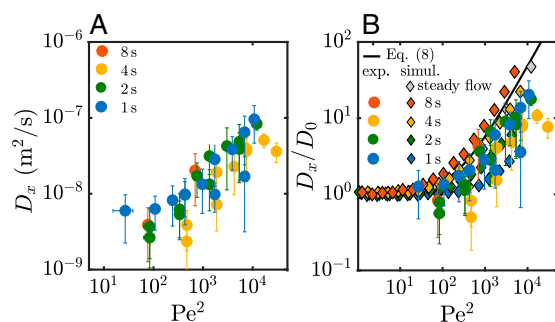


Fig. 4. Dispersion of microalgae under oscillatory microfluidic flow. (A) Experimental ensemble-averaged dispersion coefficient D_x parallel to the flow as a function of the squared Péclet number Pe^2 , for various flow-oscillation periods T , as indicated. (B) Ratio D_x/D_0 of the parallel dispersion coefficient to the raw effective diffusion coefficient as a function of squared Péclet number Pe^2 , for various flow-oscillation periods T , as indicated. The circles correspond to experimental data while the diamonds correspond to the results from two-dimensional Langevin-like simulations (21) in oscillatory flows, including one steady-flow reference case. The solid black line corresponds to Eq. 8 with $\alpha = 1/210$. Each point represents an ensemble average over trajectories longer than 20 s. Vertical error bars denote one SD, while horizontal error bars account for uncertainties in the applied pressure (as specified by the manufacturer) and in the measurement of the baseline diffusion coefficient D_0 (SI Appendix).

0.1% of the set point for pressures above 35 mbar. To maintain this condition throughout the modulation cycle, the mean pressure is set to $P_H \approx 60$ mbar, to compensate the one imposed by a hydrostatic reservoir at the opposite end of the channel. This second reservoir consists of a water column with a large free surface, and acts as a passive constant-pressure sink. Hence, the pressure is sinusoidal with mean value P_H , while the oscillation amplitude $\underline{P}$ and period T serve as independent experimental parameters. In all experiments, the pressure amplitude is set such that $\underline{P} < 10$ mbar. The Reynolds number then satisfies $Re = \frac{P}{R_h} \cdot \frac{W}{\nu} \leq 0.45$, where ν is the kinematic viscosity of water, W is the channel width, and $R_h \approx (3.1 \pm 0.2) \times 10^5$ Pa·s/m is the reduced hydraulic resistance, *i.e.* the hydraulic resistance multiplied by the cross-section area HW , measured before any experiment. Experimentally, it is not possible to exactly compensate the hydrostatic pressure, thus a small steady flow persists. The associated flow velocity is estimated from first-moment analysis on tracer trajectories to be about $10 \mu\text{m/s}$, leading to an absolute error on the Péclet number of about 2, which is much smaller than the range of this study.

Imaging and Data Acquisition. The algae are first introduced into the microfluidic channel using a slow steady flow, before the oscillating flow is turned on. The channel is mounted on an inverted microscope equipped with a $\times 5$ magnification objective (Nikon CFI TU Plan Fluor BD). *Chlamydomonas* exhibit phototactic responses (12, 64) when illuminated with visible light, particularly in the blue-green wavelength range. To prevent this effect, they are illuminated with red-filtered light (spectrum provided in *SI Appendix*). Observations are made along the vertical direction z , averaging all motion along this direction. In practice, we track horizontal motion within a $4 \text{ mm} \times 180 \mu\text{m}$ surface using the trackpy algorithm. Cell trajectories are recorded for 200 s at 40 frames per second using an Andor Neo 5.5 sCMOS camera, storing data in image stack format. Only trajectories that remain within the field of view during a reasonable time (~ 20 s) are included in the analysis. For each experiment, we fix the flow period and systematically vary the pressure amplitude, and thus the Péclet number. A no-flow reference is recorded both before and after each experimental series. In all cases, we verified that repeated measurements do not significantly affect the motility of the algae (*SI Appendix*). To minimize adaptation or memory effects between experiments, the algae are kept in the dark at zero flow for approximately 30 min before each new series.

Numerical Simulations. In addition, we perform classical Langevin-like numerical simulations of active particles (21) in two dimensions, using a forward-Euler discretization scheme, but with the inclusion of an oscillatory Poiseuille flow.

Data, Materials, and Software Availability. Numerical simulation and data analysis codes have been deposited in https://github.com/EMetBrown-Lab/Chlamy_taylor_aris (66).

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Author affiliations: ^aUniversity of Bordeaux, CNRS, LOMA, UMR 5798, Talence F-33400, France

Author contributions: M.L., Y.A., A.A., and T.S. designed research; M.L., J.L., G.d.T., and A.B. performed research; M.L., J.L., G.d.T., A.B., and Y.A. contributed new reagents/analytic tools; M.L., J.L., G.d.T., Y.A., A.A., and T.S. analyzed data; M.L., J.L., G.d.T., A.B., Y.A., A.A., and T.S. wrote the paper; and T.S. secured funding.

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